
TICVAI

INVENTORY & PROCUREMENT

Backend Structure Module

FUNCTIONAL REFERENCE DOCUMENT

Professional Edition

Prepared for review and implementation reference

TICVAI

Powered by Intelligence

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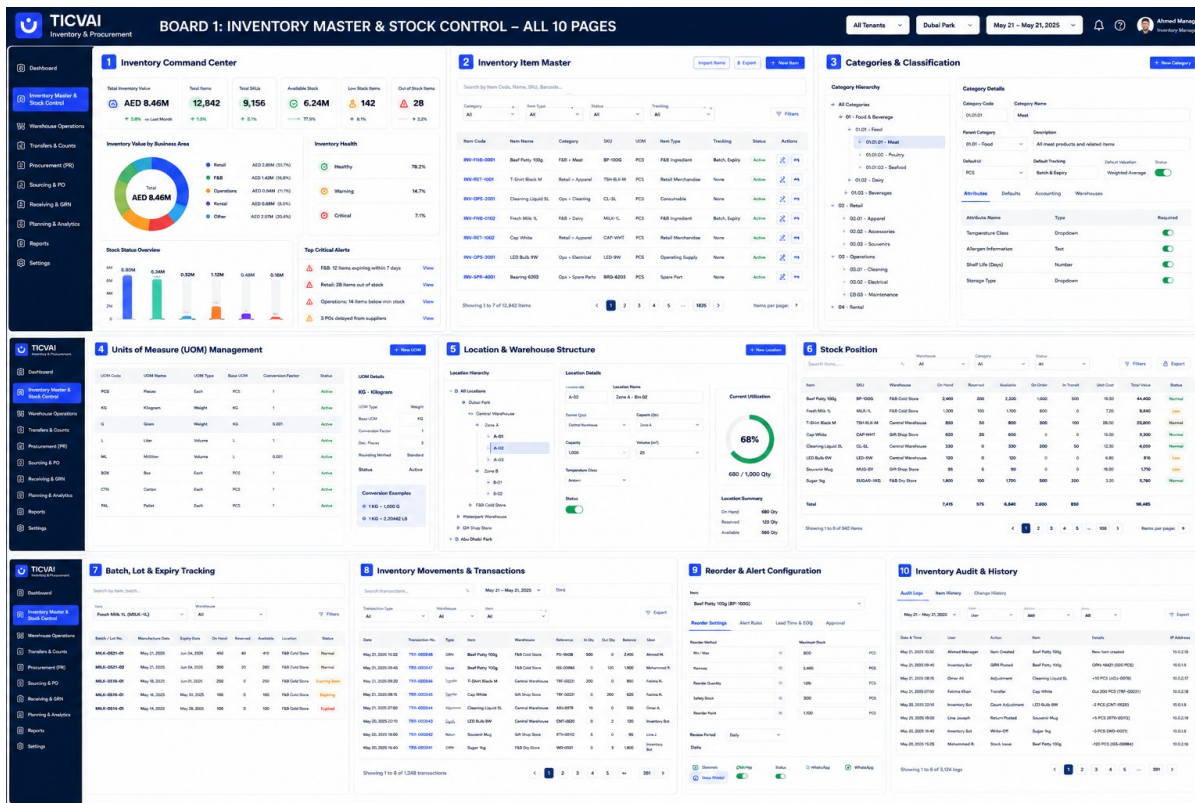
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TICVAI INVENTORY & PROCUREMENT — BOARD 1

Inventory Master & Stock Control — Scope of Work



1. Purpose

The Inventory Master & Stock Control Board shall provide the centralized enterprise inventory foundation for TICVAI.

It shall maintain a **single source of truth** for inventory items, SKU references, stock states, locations, quantities, batches, lots, serial numbers, expiry, reorder parameters, movement history, and stock control rules across all TICVAI modules.

This Board shall be consumed by:

- Food & Beverage
- Retail
- Rental
- Resource Management
- Operations
- Event Operations
- Warehouses
- Procurement
- Finance
- Employee App
- B2C / Guest App where applicable
- Reporting & AI

The core inventory model shall follow:

Inventory Item → Category → Attributes → UOM → Tracking Method → Stock Location → Quantity → Stock Status → Batch/Lot/Serial → Availability → Movement → Cost → Audit

The dedicated Inventory matrix requires centralized item management, categories, SKU/barcode management, configurable product attributes and variants, UOM management, real-time inventory, multi-location/multi-warehouse stock, available/reserved/on-order/in-transit/damaged quantities, movements, adjustments, returns, write-offs, stock counts, reorder parameters, alerts and batch/lot/serial/expiry tracking.

PAGE 1 — Enterprise Inventory Command Center

Purpose

Provide Inventory, Warehouse, Finance and Operations management with a real-time consolidated view of inventory across the entire TICVAI ecosystem.

Global Filters

- Tenant
- Company
- Business Unit
- Park
- Venue
- Warehouse
- Location
- Inventory Category
- Module
- Stock Status
- Date

KPI Cards

- Total Inventory Value
- Total Inventory Items
- Total SKUs
- Units On Hand
- Available Stock
- Reserved Stock
- On-Order Stock
- In-Transit Stock
- Damaged / Blocked Stock
- Low-Stock Items
- Out-of-Stock Items
- Overstock Items
- Expiring Soon
- Pending Adjustments
- Inventory Alerts

Inventory by Business Area

Example:

Business Area	Inventory Value	Items	Available	Reserved	Health
Retail	AED 2.85M	4,842	92%	5%	Healthy
F&B	AED 1.42M	1,864	88%	4%	Warning
Rental	AED 680K	742	84%	11%	Healthy

Business Area	Inventory Value	Items	Available	Reserved	Health
Operations	AED 940K	1,120	91%	3%	Healthy

Inventory Health

TICVAI shall identify:

- Low Stock
- Stock Shortage
- Overstock
- Expiring Inventory
- Negative Stock
- Batch Issues
- Serial Issues
- Inventory Sync Problems
- Unapproved Adjustments

The source matrix explicitly requires overstock, shortage and expiry alerts.

Quick Actions

+ Inventory Item | Stock Lookup | Adjustment | Movement History | Batch / Serial Lookup | View Alerts

PAGE 2 — Inventory Item Master

Purpose

Provide a centralized master record for every inventory-controlled item in TICVAI.

The Inventory Item Master shall be separate from a Retail sellable product or F&B menu item.

For example:

Retail Product: TICVAI T-Shirt

↓

SKU: Black / Medium

↓

Inventory Item: INV-RET-TSH-BLK-M

or:

F&B Menu Product: Classic Burger

↓

Recipe

↓

Inventory Items: Beef Patty, Bun, Cheese, Sauce.

General Information

Capture:

- Inventory Item Name
- Inventory Item Code
- SKU Reference
- Barcode
- Item Type
- Category
- Subcategory
- Description

- Brand
- Manufacturer
- Supplier Reference
- Country of Origin
- Status

Item Types

Examples:

- Retail Merchandise
- F&B Ingredient
- F&B Prepared Item
- Consumable
- Operating Supply
- Spare Part
- Rental Stock
- Equipment Consumable
- Packaging Material
- Uniform
- Cleaning Material
- General Inventory

Item types shall be configurable.

Inventory Behavior

Configure:

- Inventory Tracked
- Batch Tracked
- Lot Tracked
- Serial Tracked
- Expiry Tracked
- Barcode Tracked
- RFID Tracked
- Negative Stock Allowed
- Reservation Allowed
- Transfer Allowed
- Replenishment Enabled

Financial Information

Where authorized:

- Standard Cost
- Current Average Cost
- Valuation Category
- Inventory Account
- Consumption Account
- Adjustment Account
- Write-Off Account

The detailed valuation logic should remain integrated with Finance rather than duplicated.

PAGE 3 — Categories, Attributes & Inventory Classification

Purpose

Provide a flexible hierarchy for grouping and controlling enterprise inventory.

The matrix explicitly requires inventory item categorization and configurable product attributes.

Example Hierarchy

Food & Beverage

→ Food

→ Meat

→ Beef

or:

Retail

→ Merchandise

→ Apparel

→ T-Shirts

or:

Operations

→ Maintenance

→ Electrical

→ Spare Parts

Classification Levels

TICVAI should support configurable structures such as:

- Division
- Department
- Category
- Subcategory
- Item Group
- Family
- Brand
- Commodity
- Usage Type

Category Defaults

A category may provide default settings for:

- UOM
- Tracking Method
- Shelf-Life Requirement
- Reorder Profile
- Approval Rules
- Accounting Mapping
- Warehouse Eligibility
- Storage Conditions

Dynamic Attributes

Example:

F&B Ingredient

- Temperature Class

- Allergen Reference
- Shelf Life
- Storage Type

Retail Apparel

- Size
- Color
- Material

Spare Part

- Equipment Type
- Manufacturer
- Part Number

The attributes should be configurable without software development.

PAGE 4 — Unit of Measure & Conversion Management

Purpose

Provide centralized management of inventory measurement and conversion rules.

The matrix explicitly requires Unit of Measure management.

UOM Types

Examples:

- Piece
- Unit
- Box
- Case
- Pack
- Pallet
- Kilogram
- Gram
- Liter
- Milliliter
- Meter
- Roll

Base UOM

Every inventory item shall have a primary inventory UOM.

Example:

Milk

Base UOM:

Liter

Purchase UOM:

Case

Recipe UOM:

Milliliter

Conversion

Example:

1 Case

=

12 Bottles

1 Bottle

=

1 Liter

Therefore:

1 Case = 12 L

UOM Rules

Support:

- Purchase UOM
- Inventory UOM
- Consumption UOM
- Sales UOM
- Transfer UOM
- Count UOM

Conversion Validation

TICVAI shall prevent invalid or ambiguous conversions.

Example:

Conversion from Case to Liter requires a pack-size definition.

Effective Dating

Where conversion rules change, the system shall retain historical rules for prior transactions.

PAGE 5 — Stock by Location & Real-Time Availability

Purpose

Provide a real-time view of inventory quantities across the entire organization.

The matrix requires real-time inventory tracking across multiple locations and multiple warehouses.

Inventory Dimensions

For every Item + Location combination, display:

- On Hand
- Available
- Reserved
- Allocated
- On Order
- In Transit
- Damaged
- Blocked
- Quarantine
- Returned
- Pending Inspection

Stock Formula

The system shall clearly distinguish:

On Hand

from

Available

from

Available-to-Promise

Example:

On Hand:

100

Reserved:

15

Blocked:

5

Available:

80

Incoming tomorrow:

40

Available-to-Promise according to rule:

120

Multi-Location View

Example:

INV-BEEF-001 — Beef Patty

Central Warehouse:

480

Main Restaurant:

72

VIP Lounge:

44

Beach Kiosk:

18

In Transit:

100

On Order:

500

Location Drill-Down

Selecting a location shall display:

- Warehouse
- Zone
- Bin
- Stock Status
- Batch
- Expiry
- Last Receipt
- Last Issue
- Last Transfer
- Last Count

Warehouse zoning and bin structures will be configured more deeply in Board 2.

PAGE 6 — Batch, Lot, Serial & Expiry Management

Purpose

Provide complete traceability for inventory requiring batch, lot, serial or expiry tracking.

The matrix explicitly requires batch-number, lot, serial-number and expiry-date tracking.

Tracking Types

An inventory item may be configured as:

- None
- Batch
- Lot
- Serial
- Batch + Expiry
- Lot + Expiry
- Serial + Warranty where applicable

Batch Record

Capture:

- Batch Number
- Supplier Batch
- Inventory Item
- Quantity
- Manufacture Date
- Receipt Date
- Expiry Date
- Supplier
- Warehouse
- Location
- Quality Status

Example

Milk

Batch:

MILK-140826-A

Received:

14 Aug 2026

Expiry:

20 Aug 2026

Quantity:

320 L

Available:

280 L

Reserved:

40 L

Serial Record

Example:

RFID Scanner

Serial:

RFID-SCN-18422

Status:

Available

Warehouse:

Central Warehouse

Bin:

TECH-A04

Inventory Status

- Available
- Reserved
- Blocked
- Quarantine
- Expired
- Recalled
- Damaged
- Consumed
- Sold

Expiry Alerts

The system shall support configurable warnings such as:

90 Days

30 Days

14 Days

7 Days

1 Day

Expired

FEFO Support

For expiry-controlled items, TICVAI should support **First Expired, First Out** allocation recommendations.

PAGE 7 — Inventory Status & Stock Control Rules

Purpose

Provide centralized configuration for how inventory may move between operational states.

The matrix requires tracking of available, reserved, on-order, in-transit and damaged inventory.

Standard Stock Statuses

I recommend supporting at minimum:

- Available
- Reserved
- Allocated
- Picked
- Packed
- In Transit
- On Order
- Inspection
- Quarantine
- Blocked
- Damaged
- Returned
- Expired
- Written Off

Status Transition Rules

Example:

Available

- Reserved
- Allocated
- Picked
- Dispatched
- In Transit
- Received
- Available

Controlled Transitions

Examples:

Damaged

cannot become

Available

without inspection/approval.

Quarantine

cannot be sold or consumed.

Expired

cannot be released for normal use.

Module Rules

Stock status behavior may differ by operational context.

Example:

Retail Returned Item

- Inspection
- Available / Damaged

F&B Returned Ingredient

→ normally not returned to available consumption unless specifically approved.

Negative Inventory

The tenant shall determine:

- Negative Stock Not Allowed
- Warning Only
- Authorized Override
- Emergency Offline Exception

All overrides shall be audited.

PAGE 8 — Inventory Movement & Transaction History

Purpose

Maintain complete traceability of every inventory movement.

The source matrix explicitly requires inventory movement history and inventory audit trails.

Movement Types

Support:

- Goods Receipt
- Transfer

- Consumption
- Sale
- Return to Stock
- Reservation
- Reservation Release
- Allocation
- Picking
- Dispatch
- Adjustment
- Write-Off
- Production Issue
- Production Receipt
- Damage
- Stock Count Adjustment

Movement Table

Time	Item	Movement	From	To	Qty	Ref
10:24	T-Shirt M	Sale	Store Stock	Issued	-1	POS-18422
10:18	Beef Patty	Consumption	Main Restaurant	Consumed	-12	F&B-18428
10:10	Water Bottle	Transfer	WH-A	Store 02	-10 0	TR-18421

Transaction Detail

Display:

- Movement ID
- Item
- Batch / Serial
- Quantity
- UOM
- Source Location
- Destination
- Previous Status
- New Status
- Cost
- User/System
- Device
- Date / Time
- Source Module
- Reference Number

Source References

Examples:

- POS Transaction
- F&B Order
- Production Run
- Transfer

- Purchase Order
- GRN
- Stock Count
- Return
- Rental Transaction

Immutable History

Posted inventory transactions should not simply be deleted.

Corrections should be made using **reversal / adjustment transactions** while retaining full history.

PAGE 9 — Reorder Parameters, Min/Max & Stock Policies

Purpose

Configure the basic stock-control policies used by replenishment and planning.

The matrix explicitly requires minimum, maximum, safety stock and reorder-point management.

Parameters

Configure by:

- Item
- Location
- Warehouse
- Store
- Outlet
- Season
- Time Period

Core Fields

- Minimum Stock
- Maximum Stock
- Safety Stock
- Reorder Point
- Reorder Quantity
- Par Level
- Lead Time
- Review Frequency

Example

Retail T-Shirt / Main Gift Shop

Minimum:

20

Maximum:

100

Safety:

15

Reorder Point:

35

Reorder Quantity:

65

Different Location Profiles

The same item may have different thresholds.

Example:

Main Gift Shop:

Par 100

Exit Kiosk:

Par 30

Central Warehouse:

Par 1,000

Policy Modes

Support:

- Manual
- Min/Max
- Reorder Point
- Fixed Quantity
- Par Stock
- Forecast Based
- AI Recommended

The advanced forecasting and AI logic will be expanded in Board 6.

Threshold Alerts

Examples:

Inventory below minimum.

Inventory exceeds maximum.

Safety stock projected to be breached in 12 hours.

PAGE 10 — Inventory Exceptions, Data Quality & AI Control Center

Purpose

Provide one centralized workspace for detecting inventory-master issues, stock-control exceptions and data-quality risks.

Exception Categories

Item Master

- Duplicate Inventory Item
- Missing UOM
- Missing Category
- Missing Cost
- Invalid Barcode
- Missing Tracking Method

Stock

- Negative Stock
- Low Stock
- Overstock
- Stock Shortage
- Reserved > On Hand
- Location Conflict

Tracking

- Batch Missing
- Serial Duplicate

- Expired Stock
- Expiry Missing
- Batch Quantity Mismatch

Integration

- POS Sync Failure
- F&B Consumption Sync Failure
- Warehouse Sync Failure
- Procurement Sync Failure

Severity

Information | Warning | High | Critical

Example

Critical — Reserved Quantity Exceeds Available Stock

Item:

INV-RET-PLUSH-L

On Hand:

18

Reserved:

22

Affected Orders:

6

Actions:

Review Reservations | Check Other Locations | Release Reservation | Create Transfer

AI — Inventory Data Quality Assistant

AI should identify configuration problems proactively.

Example:

14 inventory items appear to be duplicates based on name, barcode and supplier reference.

Suggested Action:

Review Potential Duplicates

Another example:

22 items have purchase UOM defined as Case but no Case-to-Base-UOM conversion.

AI — Stock Risk Detection

Even at Board 1 level, TICVAI should identify immediate stock-control risks.

Example:

Beef Patty

Current Available:

72

Reserved / Forecast Immediate Demand:

64

Incoming:

0

Result:

High shortage risk

Recommendation:

Review other locations for internal redistribution.

The actual demand forecasting engine will be handled in Board 6.

AI — Expiry Risk

Example:

Milk Batch MILK-140826-A

Available:

180 L

Expires:

Tomorrow

Expected Consumption:

96 L

Potential Expiry:

84 L

Recommendation:

- Prioritize FEFO consumption
- Transfer to higher-demand outlet
- Review production requirement

ITEM MASTER GOVERNANCE

The Inventory Item Master should support governance throughout its lifecycle.

Lifecycle**Draft**

→ Review

→ Approved

→ Active

→ Suspended

→ Obsolete

→ Archived

Validation Before Activation

TICVAI should check:

- ✓ Unique Item Code
- ✓ Category
- ✓ Base UOM
- ✓ Inventory Behavior
- ✓ Tracking Method
- ✓ Accounting Mapping where required
- ✓ Barcode where required
- ✓ Storage Profile where required

Versioning

Significant master-data changes shall retain:

- Previous Value
- New Value
- Changed By
- Approved By
- Effective Date
- Reason

BARCODE / QR / RFID IDENTIFICATION

The Inventory Item Master shall support identification methods required across F&B and Retail.

Barcode

- Internal Barcode
- Supplier Barcode
- EAN
- UPC
- Code 128

QR

May be used for:

- Inventory Lookup
- Asset/Item Identification
- Mobile Operations

RFID

Where applicable:

- EPC
- RFID Tag Type
- Read Status

The matrix also requires barcode management at item-master level.

MULTI-TENANT & MULTI-COMPANY CONTROL

Inventory records must respect the TICVAI enterprise hierarchy.

An item may be:

Global to Tenant

or restricted to:

- Company
- Business Unit
- Park
- Venue
- Warehouse
- Module

Example:

Water Bottle

Global Item:

Yes

Used By:

Retail ✓

F&B ✓

Events ✓

This is important because TICVAI should not create separate "Retail Water Bottle" and "F&B Water Bottle" inventory records if they represent the same physical stock item.

SHARED ITEM VS MODULE PRODUCT

This distinction shall be explicit.

Inventory Item

Represents the **physical stock-controlled object**.

Examples:

Water Bottle

Beef Patty

T-Shirt Black/M

Cleaning Chemical

Retail Product

Represents something sold to the guest.

F&B Product

Represents a prepared menu item.

Procurement Item

References the same Inventory Item for purchasing.

Therefore:

One Inventory Item

may connect to:

Retail SKU

and/or

F&B Recipe

and/or

Procurement Supplier Catalog

without duplicate stock records.

APPROVAL ENGINE INTEGRATION

Board 1 shall use TICVAI's shared Approval Engine.

Examples:

New Inventory Item

→ Inventory Master Data Owner

Tracking Method Change

→ Inventory Manager

UOM Conversion Change

→ Inventory Manager / Finance where required

Negative Stock Override

→ Authorized Role

Reorder Policy Change

→ Inventory Manager

No separate approval mechanism shall be built solely for Inventory.

AUDIT REQUIREMENTS

The system shall maintain complete audit history for:

- Item Creation
- Item Modification
- Status Change
- Category Change
- UOM Change
- Barcode Change
- Batch Change
- Serial Change

- Stock Status Override
- Reorder Parameter Change
- Inventory Adjustment

Audit shall record:

User + Date/Time + Location + Action + Previous Value + New Value + Reason + Approval

BOARD 1 — FINAL 10 SCREENS

#	Screen
1	Enterprise Inventory Command Center
2	Inventory Item Master
3	Categories, Attributes & Inventory Classification
4	Unit of Measure & Conversion Management
5	Stock by Location & Real-Time Availability
6	Batch, Lot, Serial & Expiry Management
7	Inventory Status & Stock Control Rules
8	Inventory Movement & Transaction History
9	Reorder Parameters, Min/Max & Stock Policies
10	Inventory Exceptions, Data Quality & AI Control Center

COMPLETE BOARD 1 FLOW

Create Inventory Item

↓

Assign:

Category + Attributes

↓

Configure:

Base UOM + Conversions

↓

Configure:

Batch / Lot / Serial / Expiry / Barcode / RFID

↓

Configure:

Stock Status Rules

↓

Assign:

Companies / Locations / Warehouses

↓

Configure:

Min / Max / Safety / Reorder Policies

↓

Validation

↓

Approval

↓

Activate

↓

Inventory transactions begin from:

Procurement / Warehouse / Retail / F&B / Rental / Operations

↓

TICVAI continuously tracks:

On Hand**Available****Reserved****On Order****In Transit****Damaged / Blocked**

↓

Every change generates:

Inventory Movement + Audit Trail

↓

Planning, Warehouse, Procurement and AI consume the same inventory truth.

ARCHITECTURAL REQUIREMENT

This Board should establish **one central TICVAI Inventory Ledger and Item Master**.

SoftLab should not create separate inventory masters for:

Retail**F&B****Rental****Warehouse****Procurement**

Instead:

Enterprise Inventory Item Master

↓

Stock Ledger

↓

Warehouse / Location / Bin

↓

Batch / Lot / Serial

↓

Quantity + Status

↓

Cost + Movement + Audit

Then operational modules consume the same inventory services.

That means, for example:

Retail sale

→ decrements the shared inventory ledger.

F&B burger sale

→ recipe consumption decrements the same shared inventory ledger.

Procurement receipt

→ increases the same shared inventory ledger.

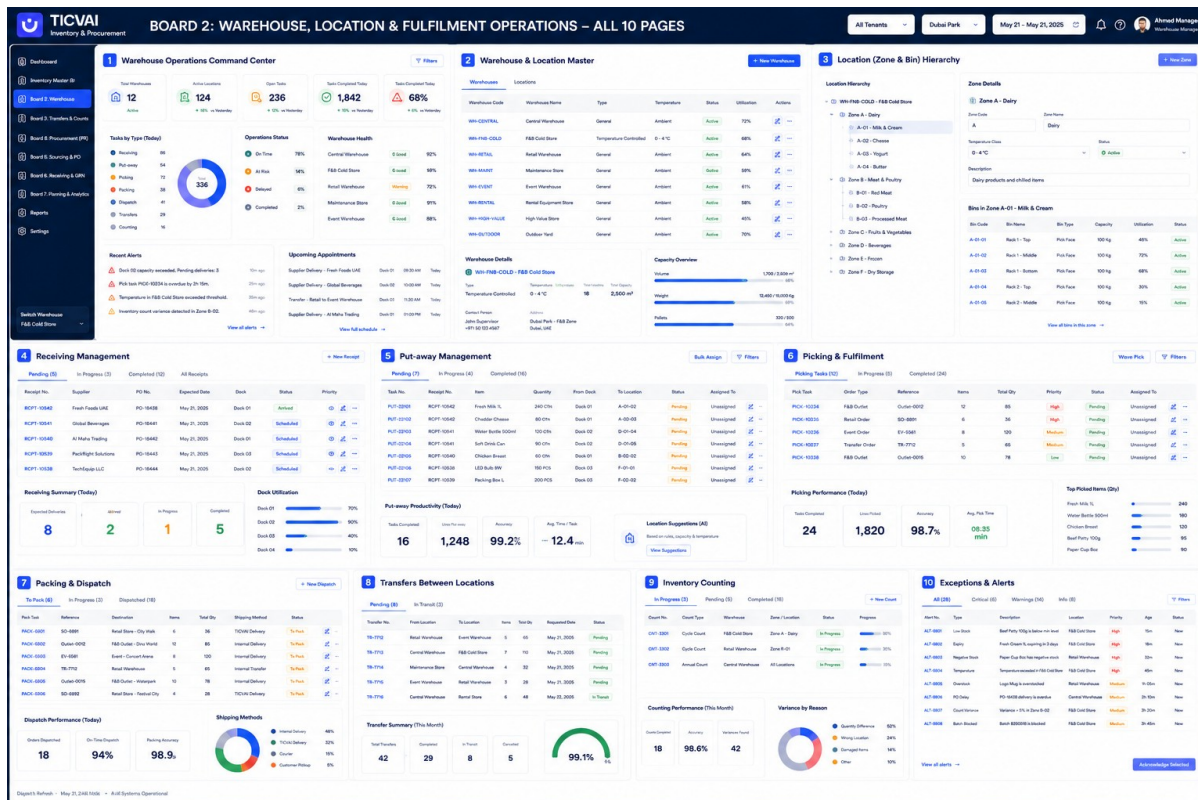
Rental allocation

→ reserves inventory or an individual tracked item in the same platform-level architecture.

This Board becomes the **foundation for every remaining Inventory & Procurement board**, and it is the most important place to prevent duplicated stock data across TICVAI.

TICVAI INVENTORY & PROCUREMENT — BOARD 2

Warehouse, Location & Fulfilment Operations — Scope of Work



1. Purpose

The Warehouse, Location & Fulfilment Operations Board shall provide TICVAI with the centralized operational layer for managing **warehouses, storage locations, zones, bins, receiving, put-away, internal warehouse movements, picking, packing, staging and dispatch.**

Board 1 established **what inventory exists and its enterprise stock position.**

Board 2 shall establish:

Where is it physically located? → How does it enter the warehouse? → Where should it be stored? → How is it picked? → How is it packed? → How does it leave?

The architecture shall follow:

Warehouse → Zone → Storage Location → Bin → Inventory Item → Batch / Serial → Stock Status → Task → Movement → Employee / Device → Confirmation

Board 2 shall support warehouse operations serving:

- Food & Beverage
- Retail
- Rental
- Operations
- Maintenance
- Events
- Temporary / Event Warehouses
- Central Distribution Centers
- Venue Stores
- Back-of-House Storage

- Cold / Frozen Storage
- Secure Storage
- Quarantine Areas

It shall use the **same Shared Inventory Ledger established in Board 1**. Warehouse operations shall create controlled inventory movements against that ledger rather than maintaining a separate warehouse stock balance.

PAGE 1 — Warehouse Operations Command Center

Purpose

Provide Warehouse and Inventory management with a real-time view of all warehouse activity, workload, capacity, inventory movement and operational exceptions.

Global Filters

- Tenant
- Company
- Business Unit
- Park
- Venue
- Warehouse
- Zone
- Operation Type
- Priority
- Status
- Shift
- Date / Time

KPI Cards

- Warehouses Active
- Inventory Value
- Warehouse Capacity Used %
- Receipts Today
- Units Received
- Pending Put-Away
- Open Picking Tasks
- Orders / Transfers Ready to Pack
- Pending Dispatch
- In-Transit Shipments
- Warehouse Tasks
- Overdue Tasks
- Receiving Exceptions
- Picking Accuracy
- On-Time Dispatch %
- Critical Alerts

Operational Pipeline

The dashboard should visually show:

Expected Receipts

→ Receiving

- **Inspection**
- **Put-Away**
- **Available Stock**
- **Picking**
- **Packing**
- **Staging**
- **Dispatch**

Example:

Expected Receipts: **18**

Receiving: **4**

Pending Put-Away: **7**

Picking: **22**

Packing: **9**

Staged: **6**

Ready to Dispatch: **8**

Warehouse Performance

Warehouse	Capacity	Receiving	Put-Away	Picking	Dispatch	Health
Central WH	78%	6	4	18	7	Healthy
F&B WH	84%	4	8	12	5	Warning
Retail WH	69%	3	2	14	6	Healthy
Event WH	92%	5	7	22	11	Critical

Operational Alerts

Examples:

F&B Cold Storage is at 94% capacity.

3 receiving tasks have exceeded SLA.

Transfer TR-18421 is awaiting picking.

14 units could not be located during picking.

Event Warehouse staging area is projected to exceed capacity at 16:30.

Quick Actions

+ **Warehouse** | **Receive Stock** | **Put-Away** | **Create Pick** | **Pack** | **Dispatch** | **Transfer** | **Task Center**

PAGE 2 — Warehouse, Zone & Bin Configuration

Purpose

Configure the physical warehouse structure used by TICVAI.

The system should support:

Warehouse

- **Area**
- **Zone**
- **Aisle**
- **Rack**
- **Shelf**
- **Bin**

Not every tenant needs every hierarchy level, so the structure should be configurable.

Warehouse Master

Capture:

- Warehouse Name
- Warehouse Code
- Company
- Business Unit
- Park
- Venue
- Address / Location
- Warehouse Type
- Operating Hours
- Manager
- Status

Warehouse Types

Examples:

- Central Warehouse
- Retail Warehouse
- F&B Warehouse
- Cold Storage
- Frozen Storage
- Dry Storage
- Rental Warehouse
- Maintenance Warehouse
- Event Warehouse
- Temporary Warehouse
- Secure Warehouse

Zone Configuration

Examples:

Central Warehouse

- Receiving Zone
- Inspection Zone
- General Storage
- High-Value Storage
- Picking Zone
- Packing Zone
- Staging Zone
- Dispatch Zone
- Returns Zone
- Quarantine Zone

Bin Master

Capture:

- Bin Code
- Zone
- Aisle

- Rack
- Shelf
- Bin Type
- Capacity
- Weight Limit
- Volume Limit
- Temperature Requirement
- Status

Bin Status

- Available
- Occupied
- Full
- Reserved
- Blocked
- Inspection
- Maintenance

Capacity Visualization

Warehouse managers should be able to see:

Warehouse → Zone → Bin Utilization

using percentage and heatmap views.

PAGE 3 — Storage Rules & Location Assignment

Purpose

Determine **where inventory is permitted or recommended to be stored**.

TICVAI should not allow arbitrary storage where operational or product rules prohibit it.

Storage Profiles

An Inventory Item may require:

- Ambient
- Chilled
- Frozen
- Dry
- Secure
- Hazard-Controlled
- High-Value
- Fragile
- Oversized
- Food-Safe
- Quarantine

Item-to-Location Rules

Example:

Fresh Milk

Storage:

Chilled

Temperature:

2°C–5°C

Eligible Zones:

Cold Storage A / Cold Storage B

Not Allowed:

Dry Storage

Location Assignment

Rules may consider:

- Item Category
- Storage Condition
- Size
- Weight
- Velocity
- Batch
- Expiry
- Hazard Classification where applicable
- Security Requirement
- Business Area
- Picking Frequency

Fixed vs Dynamic Bins

Support:

Fixed Bin

Item normally stored in designated location.

Dynamic Bin

System selects available suitable storage.

Multiple Items per Bin

Configurable:

- Single SKU Only
- Multiple SKU Allowed
- Same Batch Only
- Mixed Batch Allowed
- Serial-Controlled

Capacity Validation

Before assigning inventory:

Current Bin Occupancy + Incoming Quantity ≤ Bin Capacity

If capacity is exceeded:

Selected bin cannot accommodate the receipt.

TICVAI shall recommend another eligible location.

PAGE 4 — Receiving Dock & Inbound Operations

Purpose

Manage inventory entering a warehouse from suppliers, transfers, production, returns or other authorized sources.

Inbound Sources

Support:

- Purchase Order
- Inter-Warehouse Transfer
- Store Return
- F&B Production
- Supplier Delivery
- Event Return
- Rental Return where inventory-controlled
- Manual Authorized Receipt

Expected Receipts

Display:

- Receipt / Reference
- Source
- Supplier
- PO / Transfer
- Expected Date
- Expected Time
- Items
- Units
- Priority
- Dock
- Status

Receiving Methods

Employees should be able to receive using:

- Barcode Scanner
- QR
- RFID
- Mobile Device
- Employee App
- Manual Entry

Receiving Validation

For each item:

Expected

vs

Received

vs

Accepted

vs

Rejected / Damaged

Example:

Expected:

100 units

Received:

98

Accepted:

96

Damaged:

2

Short:

2

Receiving Exceptions

Support:

- Short Delivery
- Over Delivery
- Wrong Item
- Wrong Variant
- Damaged
- Expired
- Insufficient Remaining Shelf Life
- Batch Mismatch
- Serial Mismatch
- Unrecognized Barcode
- Unplanned Delivery

Dock Assignment

Where applicable:

- Dock Number
- Appointment
- Vehicle
- Supplier
- Arrival Time
- Start Receiving
- Complete Receiving

PAGE 5 — Inspection, Quality Hold & Put-Away

Purpose

Control inventory after physical receipt and before it becomes available for operational use.

Receipt Flow

Received

↓

Inspection Required?

No

→ Put-Away

Yes

→ Inspection

→ Accepted / Rejected / Quarantine

→ Put-Away

Inspection Information

Capture:

- Receipt
- Item

- Batch / Serial
- Quantity
- Inspector
- Inspection Type
- Condition
- Result
- Notes
- Attachments / Photos where required

Inspection Results

- Accepted
- Accepted with Exception
- Rejected
- Quarantine
- Damaged

Put-Away Task

Example:

PA-18428

Item:

Fresh Milk 1L

Quantity:

240

Current Location:

Receiving Dock 02

Recommended Location:

Cold Storage A / Rack 03 / Bin 04

Priority:

High

Reason:

Expiry-Controlled / FEFO

Put-Away Strategies

Support:

- Fixed Bin
- Nearest Available Bin
- Same Item Consolidation
- Same Batch Consolidation
- Empty Bin
- Capacity Optimized
- Velocity Based
- FEFO Compatible
- AI Recommended

Confirmation

Employee scans:

Item



Source Location



Destination Bin



Quantity



Confirm

The shared inventory ledger updates the item's physical location.

PAGE 6 — Warehouse Task & Workforce Management

Purpose

Create and manage warehouse operational tasks and assign them to employees or teams.

Task Types

- Receiving
- Inspection
- Put-Away
- Replenishment
- Picking
- Packing
- Staging
- Dispatch
- Internal Movement
- Stock Count
- Cycle Count
- Return Handling

Task Fields

- Task Number
- Task Type
- Warehouse
- Zone
- Priority
- Item / Order
- Quantity
- Source
- Destination
- Assigned Employee
- Team
- Created Time
- SLA
- Status

Task Status

New

- Assigned
- Accepted
- In Progress
- Completed

or

- Exception

Priority

- Low
- Normal
- High
- Urgent
- Critical

Task Queue

Example:

Task	Type	Zone	Assigned	Priority	SLA	Status
PA-18428	Put-Away	Cold A	Omar A.	High	12 min	In Progress
PK-18431	Pick	Retail	Sara K.	Urgent	8 min	Assigned
RC-18434	Receive	Dock 2	Team B	Normal	24 min	In Progress

Workload Visibility

Supervisor should see:

- Employee
- Active Tasks
- Completed Tasks
- Pending Tasks
- Average Completion Time
- Current Zone

Resource Management Integration

Employee availability, roster and workload should integrate with the common **TICVAI Resource Management Engine**, rather than Board 2 maintaining a separate employee scheduling system.

PAGE 7 — Picking & Replenishment Operations

Purpose

Manage warehouse picking for stores, F&B outlets, events, internal requests and fulfilment operations.

Pick Sources

Picking may originate from:

- Retail Replenishment
- F&B Replenishment
- Store Transfer
- Warehouse Transfer
- Event Requirement

- Internal Department Request
- Customer Fulfilment where applicable
- Production Requirement

Pick List

Capture:

- Pick Number
- Source Request
- Warehouse
- Destination
- Required Time
- Priority
- Items
- Quantities
- Assigned Picker
- Status

Picking Strategies

Support:

- Single Order Picking
- Batch Picking
- Wave Picking
- Zone Picking
- Priority Picking
- FEFO
- FIFO
- Serial-Specific Picking

Pick Sequence

TICVAI should optimize:

Bin 01 → Bin 04 → Bin 08 → Bin 12

instead of unnecessary warehouse movement.

Scan Validation

Picker scans:

Location

↓

Item

↓

Batch / Serial

↓

Quantity

↓

Confirm

Picking Exception

Examples:

- Item Not Found
- Insufficient Quantity
- Wrong Batch

- Damaged Item
- Barcode Mismatch
- Serial Mismatch
- Location Empty

Short Pick

Example:

Requested:

100

Picked:

94

Short:

6

Actions:

Search Alternate Bin | Substitute where allowed | Partial Fulfilment | Escalate

PAGE 8 — Packing, Staging & Dispatch

Purpose

Manage outbound preparation after inventory has been picked.

Packing Workspace

Display:

- Pick Number
- Destination
- Items
- Picked Quantity
- Packing Status
- Package Count
- Weight
- Volume
- Special Handling

Packing Validation

The employee verifies:

Picked Item

vs

Required Item

before closing the package.

Package Information

Capture:

- Package ID
- Container Type
- Weight
- Dimensions
- Items
- Quantity
- Destination

- Barcode / QR
- Seal Number where applicable

Staging

After packing:

Package

→ assigned to

Staging Area

based on:

- Destination
- Route
- Dispatch Time
- Vehicle
- Priority

Dispatch

Capture:

- Dispatch Number
- Destination
- Vehicle
- Driver / Courier
- Packages
- Departure Time
- Expected Arrival
- Status

Dispatch Lifecycle

Ready

→ Staged

→ Loaded

→ Dispatched

→ In Transit

→ Received

Dispatch Verification

Before departure:

- ✓ All Packages Loaded
- ✓ Correct Destination
- ✓ Required Documents
- ✓ Temperature Requirements where applicable
- ✓ Seal Verification where applicable

PAGE 9 — Internal Warehouse Movement & Bin Replenishment

Purpose

Manage stock movement **inside the same warehouse** without confusing it with warehouse-to-warehouse transfers.

Movement Examples

- Reserve Storage → Picking Bin

- Bin → Bin
- Receiving → Inspection
- Inspection → Storage
- Storage → Staging
- Damaged → Quarantine
- Returns → Inspection

Internal Movement

Capture:

- Movement Number
- Item
- Batch / Serial
- Quantity
- Source Bin
- Destination Bin
- Reason
- Employee
- Priority

Picking-Bin Replenishment

Example:

Picking Bin RET-A01

Available:

8

Minimum:

20

Reserve Storage:

240

Recommended Replenishment:

42 units

Replenishment Trigger

Support:

- Minimum Level
- Fixed Schedule
- Pick Demand
- Wave Requirement
- Event Requirement
- AI Recommendation

Bin Consolidation

TICVAI may identify inefficient storage.

Example:

Same SKU exists in:

Bin A01:

8 units

Bin A04:

12 units

Bin B06:

6 units

Recommendation:

Consolidate into Bin A04 to release two storage locations.

Movement Validation

System shall validate:

- Destination Capacity
- Storage Compatibility
- Batch Rules
- Status Rules
- Serial Rules

before confirmation.

PAGE 10 — Warehouse Exceptions, AI Optimization & Control Center

Purpose

Provide a centralized operational workspace for warehouse exceptions, bottlenecks and optimization recommendations.

Exception Categories

Receiving

- Late Delivery
- Receiving SLA Exceeded
- Quantity Variance
- Damaged Receipt
- Unexpected Delivery

Put-Away

- Pending Put-Away
- No Eligible Bin
- Capacity Conflict
- Wrong Storage Condition

Picking

- Short Pick
- Item Not Found
- Wrong Item
- Batch Mismatch
- Serial Mismatch
- Pick SLA Exceeded

Packing

- Packing Difference
- Missing Item
- Package Weight Difference

Dispatch

- Dispatch Delayed
- Missing Package
- Vehicle Delay

- Receipt Confirmation Missing

Capacity

- Warehouse Near Capacity
- Zone Full
- Bin Over Capacity
- Staging Congestion

Severity

Information | Warning | High | Critical

AI — SMART PUT-AWAY

TICVAI should recommend the most appropriate storage location.

Example:

Item

Water Bottle 500ml

Quantity:

1,200

Candidate Locations

A01:

Available Capacity 1,500

Distance to Picking:

12m

A08:

Capacity 3,000

Distance:

38m

B04:

Capacity 2,000

Distance:

24m

AI Recommendation

A01

Reason:

- High Sales Velocity
- Near Picking Area
- Sufficient Capacity
- Compatible Storage
- Expected frequent replenishment

Actions:

Accept | Select Alternative | View Analysis

AI — PICK PATH OPTIMIZATION

TICVAI should optimize warehouse movement.

Example:

Standard Route:

420 meters

Optimized Route:

286 meters

Reduction:

31.9%

Expected Pick Time:

18 min → 13 min

The recommendation should consider:

- Warehouse Layout
- Pick Locations
- Item Sequence
- Priority
- Congestion where available
- Picker Zone

AI — WAREHOUSE CAPACITY FORECAST

TICVAI should forecast warehouse utilization using:

- Current Inventory
- Expected Purchase Receipts
- Expected Transfers
- Event Demand
- Seasonal Inventory
- Outbound Forecast
- Existing Capacity

Example:

Event Warehouse is currently 78% utilized and projected to reach 96% on Friday at 14:00 following three scheduled supplier receipts.

Recommended Actions:

1. Move slow-moving inventory to Central Warehouse.
2. Expedite scheduled outbound transfers.
3. Redirect eligible receipt to alternate warehouse.

AI — TASK PRIORITIZATION

Instead of treating all warehouse tasks equally, TICVAI should calculate operational urgency.

Example:

Task A

Retail replenishment

Stock-Out Risk:

High

Guest Revenue Impact:

AED 12K

Required:

45 minutes

Task B

General Put-Away

Stock-Out Risk:

None

Required:

4 hours

Recommendation:

Prioritize Task A

The user may accept the recommendation or follow configured priority rules.

AI — BOTTLENECK DETECTION

TICVAI should identify operational congestion.

Example:

Receiving Dock

Inbound:

8 shipments

Available Receiving Teams:

2

Average Processing:

34 min

Projected Delay:

76 minutes

Recommendation:

Temporarily assign one additional warehouse employee to receiving.

This recommendation should connect with the Resource Management module.

AI — SLOT OPTIMIZATION

TICVAI should analyze item velocity and warehouse location.

Example:

High-Velocity Items Stored Far From Picking

1. Water Bottle
2. Logo T-Shirt
3. Packaging Bag

Estimated unnecessary walking:

18.4 km/week

Recommendation:

Relocate these items to the primary picking zone.

Expected:

Picking Time:

-11%

Warehouse Travel:

-16%

MOBILE / EMPLOYEE APP INTEGRATION

Board 2 should be heavily integrated with the TICVAI Employee App.

Warehouse employees should be able to:

- View Assigned Tasks
- Scan Barcode
- Scan QR
- Scan RFID where supported
- Receive Goods
- Record Quantity
- Record Damage
- Capture Photo

- Confirm Inspection
- Perform Put-Away
- Pick Inventory
- Report Short Pick
- Pack
- Stage
- Dispatch
- Move Stock
- Count Stock
- View Item Location
- Search Inventory

Example Employee Task

Urgent Pick — PK-18431

Destination:

Main Gift Shop

Required:

10:45

Items:

8 SKUs / 64 Units

Start:

Begin Picking

BARCODE / QR / RFID OPERATIONS

Scanning should be part of the core warehouse architecture, not an optional afterthought.

Receiving

Scan:

Shipment → Item → Batch / Serial → Quantity

Put-Away

Scan:

Item → Destination Bin

Picking

Scan:

Bin → Item → Quantity

Packing

Scan:

Item → Package

Dispatch

Scan:

Package → Vehicle / Dispatch

Stock Movement

Scan:

Source → Item → Destination

This reduces manual entry and improves inventory accuracy.

OFFLINE / CONNECTIVITY HANDLING

Warehouse mobile workflows should support controlled offline behavior where required.

When connectivity is temporarily unavailable:

- Previously assigned tasks remain accessible where supported.
- Scans may be locally queued.
- Transactions should receive temporary device references.
- Synchronization occurs when connectivity returns.
- Conflicts shall be surfaced for resolution.

Example:

Item was moved by another user while device was offline.

TICVAI should not silently overwrite the newer inventory state.

F&B INTEGRATION

Example:

Main Restaurant requires:

Beef Patty:

120

Cheese:

80

Buns:

120

F&B Board creates replenishment requirement.

↓

Shared Inventory validates stock.

↓

Board 2 generates warehouse pick.

↓

Warehouse picks.

↓

Packs / Stages.

↓

Dispatches.

↓

Restaurant receives.

↓

Shared inventory locations update.

RETAIL INTEGRATION

Example:

Retail Board 4 identifies:

Main Gift Shop projected T-Shirt shortage.

↓

Creates replenishment request.

↓

Board 2 creates:

Pick Task

↓

Warehouse picks:

100 units



Packs



Dispatches



Store receives



Shared Inventory updates.

PROCUREMENT INTEGRATION

Example:

Purchase Order:

PO-18421



Supplier shipment arrives.



Board 2:

Receiving



Inspection



GRN / Receipt Process



Put-Away



Shared Inventory increases.

The procurement and financial matching side will be covered in later Boards.

RESOURCE MANAGEMENT INTEGRATION

Warehouse employees and operational resources should come from the shared Resource Engine.

Example:

Board 2 forecasts:

22 picking tasks between 14:00–16:00

Current Warehouse Staff:

5

Required:

8

Gap:

3 Employees

TICVAI can recommend additional staff through Resource Management rather than maintaining duplicate warehouse rosters.

APPROVAL ENGINE INTEGRATION

Board 2 shall use TICVAI's shared Approval Engine where approval is required.

Examples:

Unexpected Receipt

→ Warehouse Supervisor

Over-Delivery Above Tolerance

→ Procurement / Inventory

Rejected High-Value Goods

→ Warehouse + Procurement

Emergency Dispatch

→ Configurable Approval

Manual Inventory Movement Override

→ Inventory Supervisor

Normal warehouse operations should not be overloaded with unnecessary approvals.

Approvals should be policy-based and exception-driven.

ROLE-BASED ACCESS**Warehouse Director**

Full warehouse visibility.

Warehouse Manager

Warehouse configuration + operations.

Supervisor

Task management + exceptions.

Receiver

Receiving + inspection according to permission.

Picker

Assigned picking tasks.

Packer

Packing / staging.

Dispatcher

Dispatch operations.

Inventory Controller

Inventory movement + reconciliation.

Access shall use TICVAI's common **RBAC/PBAC framework**.

AUDIT & TRACEABILITY

Every warehouse operation shall retain:

- Transaction / Task ID
- Inventory Item
- Batch / Serial
- Quantity
- Source
- Destination
- Employee
- Device
- Date / Time
- Reference
- Previous State
- New State
- Exception
- Approval where applicable

Management should be able to trace:

Supplier

- Receipt
- Batch
- Warehouse
- Zone
- Bin
- Pick
- Package
- Dispatch
- Destination

BOARD 2 — FINAL 10 SCREENS

#	Screen
1	Warehouse Operations Command Center
2	Warehouse, Zone & Bin Configuration
3	Storage Rules & Location Assignment
4	Receiving Dock & Inbound Operations
5	Inspection, Quality Hold & Put-Away
6	Warehouse Task & Workforce Management
7	Picking & Replenishment Operations
8	Packing, Staging & Dispatch
9	Internal Warehouse Movement & Bin Replenishment
10	Warehouse Exceptions, AI Optimization & Control Center

COMPLETE BOARD 2 OPERATIONAL FLOW

INBOUND

Supplier / Transfer / Production / Return



Inspection Required?

Yes

- Inspection
- Accept / Reject / Quarantine

No

→ Continue



Put-Away Recommendation



Scan Destination



Shared Inventory Location Updated

**Available Stock**

INTERNAL OPERATIONS**Available Stock**

Demand from:

Retail / F&B / Event / Rental / Operations

Warehouse Task



Pick



Scan Validation



Pack



Stage



Dispatch

**In Transit**

Destination Receipt

Shared Inventory Updated

CONTINUOUS OPTIMIZATION

Warehouse activity



AI analyzes:

Capacity + Tasks + Demand + Item Velocity + SLAs + Locations

Detect:

Congestion / Inefficient Slots / Delays / Short Picks / Capacity Risk

Recommend:

Reprioritize Tasks / Change Bin / Optimize Route / Reallocate Staff / Redirect Stock

Human Approval where required



Execute



Measure Result

IMPORTANT ARCHITECTURAL BOUNDARY

There is one point I recommend we make very clear to SoftLab.

Board 2 is the Warehouse Execution layer.

It should **not duplicate Board 1's Inventory Ledger**, and it should **not duplicate Procurement**.

The architecture should be:

Board 1 — Inventory Master & Stock Ledger

provides:

Item + Quantity + Status + Batch + Serial + Cost + Movement

**Board 2 — Warehouse Execution**

provides:

Warehouse + Zone + Bin + Receive + Put-Away + Pick + Pack + Stage + Dispatch



Later:

Procurement Boards

provide:

Supplier + PR + RFQ + Quote + PO + Supplier Delivery

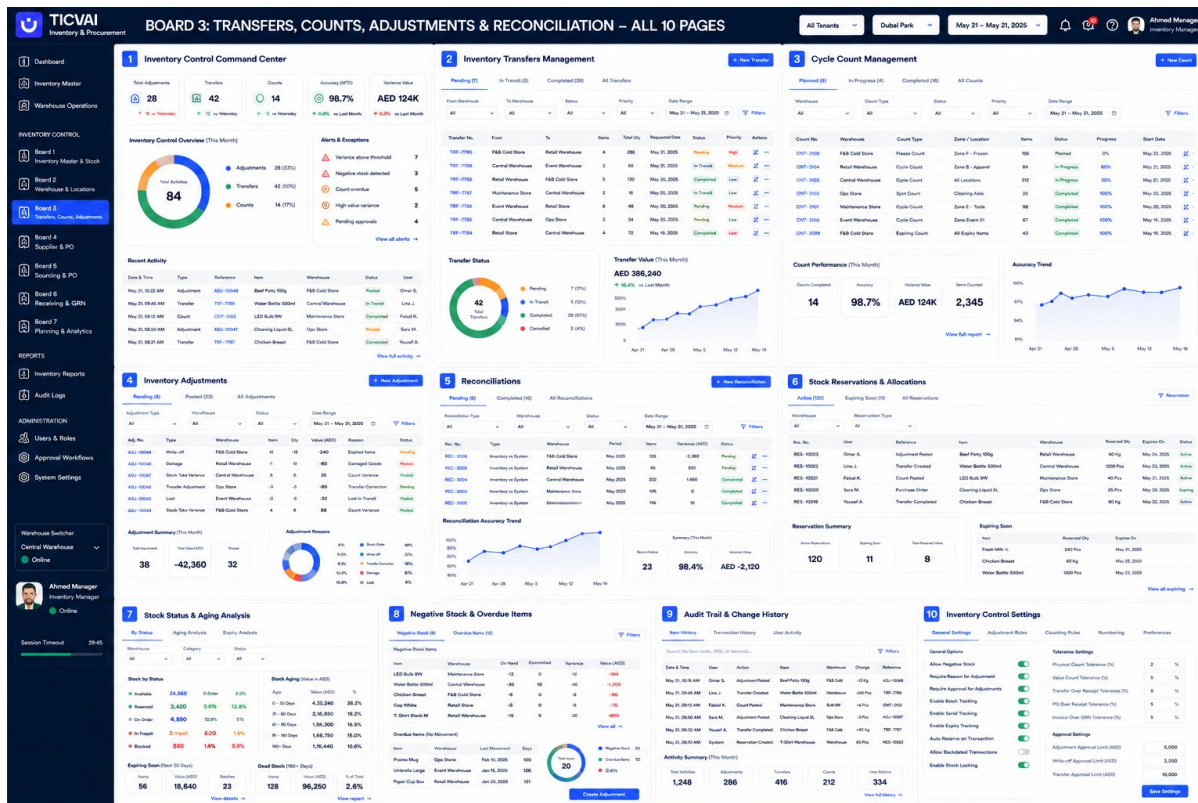
Therefore, when a warehouse employee moves 50 units from **Receiving Dock 01 → Bin A-04**, Board 2 manages the physical workflow, but the transaction posts into the **same Shared Inventory Ledger from Board 1**.

This distinction will prevent SoftLab from accidentally building a separate Warehouse Management System with its own inventory balance.

It also gives us the foundation for **Board 3 — Transfers, Counts & Reconciliation**, where we should handle enterprise transfers, physical/cycle counts, stock variance, adjustments and reconciliation across all locations.

TICVAI INVENTORY & PROCUREMENT — BOARD 3

Transfers, Stock Counts, Adjustments & Reconciliation — Scope of Work



1. Purpose

The Transfers, Stock Counts, Adjustments & Reconciliation Board shall provide TICVAI with the centralized control layer for **moving inventory between locations, verifying physical stock, identifying discrepancies, approving adjustments, reconciling inventory balances, and maintaining complete stock accountability.**

Board 1 established:

What inventory exists and how much stock TICVAI has.

Board 2 established:

Where inventory is physically stored and how warehouse operations execute receiving, put-away, picking, packing and dispatch.

Board 3 shall establish:

How stock moves between locations → How physical stock is verified → How discrepancies are investigated → How corrections are controlled → How the inventory ledger remains accurate.

The core architecture shall follow:

Transfer / Count Request → Inventory Item → Source Location → Destination / Physical Quantity → Variance → Investigation → Approval → Adjustment → Inventory Ledger → Finance Impact → Audit

Board 3 shall use the **same Shared Inventory Ledger from Board 1** and the warehouse/location structure from Board 2.

PAGE 1 — Inventory Control Command Center

Purpose

Provide Inventory Controllers and management with a consolidated view of stock transfers, physical counts, variances, adjustments and reconciliation status.

Global Filters

- Tenant
- Company
- Business Unit
- Park
- Venue
- Warehouse
- Store / Outlet
- Inventory Category
- Item / SKU
- Transfer Status
- Count Status
- Variance Status
- Date / Period

KPI Cards

- Transfers Today
- Transfers In Transit
- Pending Receipts
- Transfer Exceptions
- Counts Scheduled
- Counts In Progress
- Counts Completed
- Items with Variance
- Total Variance Value
- Pending Adjustments
- Shrinkage Value
- Unexplained Variance
- Reconciliation Exceptions
- Critical Alerts

Transfer Snapshot

Example:

Requested: 42

Approved: 35

Picking: 8

In Transit: 14

Awaiting Receipt: 7

Completed: 126

Count Snapshot

Scheduled Today: 18

In Progress: 6

Completed: 9

Variance Review: 3

Inventory Accuracy

Example:

98.7%

Target:

99.5%**Critical Alerts**

Examples:

Transfer TR-18425 has been in transit for 5 hours beyond expected arrival.

Physical count identified AED 8,420 variance in Retail Warehouse.

14 serialized items require reconciliation.

F&B Cold Store variance exceeds configured tolerance.

Quick Actions**+ Transfer | + Stock Count | Cycle Count | Adjustment | Reconciliation | Variance Review**

PAGE 2 — Inventory Transfer Management

Purpose

Create, approve, execute and monitor inventory transfers between TICVAI inventory locations.

Transfer Types

Support:

- Warehouse → Warehouse
- Warehouse → Store
- Warehouse → F&B Outlet
- Store → Store
- Outlet → Outlet
- Store / Outlet → Warehouse
- Venue → Venue
- Park → Park
- Event Warehouse → Venue
- Return to Central Warehouse
- Emergency Transfer

Transfer Header

Capture:

- Transfer Number
- Transfer Type
- Source
- Destination
- Requested Date
- Required Date / Time
- Priority
- Requesting Department
- Requested By
- Reason
- Status

Transfer Items

For each line:

- Inventory Item

- SKU
- Batch / Lot
- Serial Number
- Requested Quantity
- Approved Quantity
- Picked Quantity
- Dispatched Quantity
- Received Quantity
- UOM

Availability Validation

Before submission, TICVAI shall validate:

On Hand

Available

Reserved

Safety Stock

Source Demand

Example

Requested:

100 Water Bottles

Source Available:

280

Source Safety Stock:

100

Available for Transfer:

180

Result:

Transfer Allowed

Transfer Lifecycle

Draft

- Submitted
- Approval where required
- Approved
- Picking
- Packed
- Dispatched
- In Transit
- Received
- Reconciled
- Completed

PAGE 3 — Transfer Dispatch, Receipt & Discrepancy Control

Purpose

Provide end-to-end control over inventory while it is moving between locations.

Dispatch Confirmation

Capture:

- Transfer Number
- Packages
- Item
- Quantity
- Batch / Serial
- Dispatch Time
- Vehicle / Courier
- Employee
- Seal Number where applicable

Once dispatched:

Source Available Stock decreases

and quantity becomes:

In Transit

Destination Receipt

Receiving location confirms:

- Item
- Quantity
- Batch
- Serial
- Condition
- Received Time
- Employee

Example

Dispatched:

100 units

Received:

98

Damaged:

1

Missing:

1

Discrepancy Status

TICVAI creates:

Transfer Variance — 2 Units

The transfer shall not simply close as 100 units received.

Discrepancy Types

- Short Receipt
- Over Receipt
- Damage in Transit
- Wrong Item
- Wrong Variant
- Batch Mismatch
- Serial Mismatch

- Package Missing
- Seal Broken
- Unknown Difference

Actions

Accept Variance | Investigate | Return Item | Create Claim | Escalate

Chain of Custody

The system should maintain:

Picked By → Packed By → Dispatched By → Transport → Received By

for controlled inventory.

PAGE 4 — Stock Count Planning & Scheduling

Purpose

Allow Inventory teams to configure and schedule physical inventory verification.

Count Types

Support:

- Full Physical Inventory
- Cycle Count
- Spot Count
- Blind Count
- Recount
- High-Value Count
- Category Count
- Bin Count
- Batch Count
- Serial Count
- Event Closing Count

Count Scope

A count may target:

- Company
- Park
- Venue
- Warehouse
- Zone
- Bin
- Store
- Outlet
- Category
- Item
- SKU
- Batch
- Serial Range

Count Schedule

Capture:

- Count Name

- Count Type
- Scope
- Scheduled Date
- Start Time
- Expected Completion
- Assigned Team
- Supervisor
- Freeze Policy
- Blind Count
- Recount Threshold

Recurring Counts

Support:

- Daily
- Weekly
- Monthly
- Quarterly
- Annual
- Custom Frequency

ABC-Based Count Frequency

Example:

A Items

Weekly

B Items

Monthly

C Items

Quarterly

Count Calendar

Managers should have:

Day | Week | Month

calendar views showing planned counts across locations.

PAGE 5 — Physical & Cycle Count Execution

Purpose

Allow warehouse/store employees to perform inventory counts accurately using mobile or desktop workflows.

Count Workflow

Employee opens:

Count Task CT-18428

↓

Select / Scan Location

↓

Scan Item

↓

Scan Batch / Serial where required

↓

Enter Quantity



Confirm

Counting Methods

Support:

- Manual Entry
- Barcode Scan
- QR Scan
- RFID
- Mobile Device
- Employee App

Blind Count

For controlled counts, the employee should not see the system quantity.

Example:

System Quantity:

Hidden

Employee Count:

84

After submission:

System:

87

Variance:

-3

Serial Count

For serialized inventory:

Expected Serials

vs

Scanned Serials

TICVAI identifies:

- Missing Serial
- Unexpected Serial
- Duplicate Scan
- Wrong Location

Count Progress

Example:

Locations:

48

Completed:

31

Progress:

64.6%

Items:

1,240

Counted:

824

Count Task Status**Scheduled**

- Assigned
 - In Progress
 - Submitted
 - Variance Review
 - Approved
 - Posted
-

PAGE 6 — Variance Investigation & Root-Cause Management

Purpose

Provide a controlled process for investigating differences between physical stock and system stock.

Variance Calculation

For each count:

Physical Quantity – System Quantity = Quantity Variance

The system shall also calculate:

Variance Value

based on the applicable inventory valuation method.

Example

SKU:

TSH-BLK-M

System:

120

Physical:

113

Variance:

-7

Unit Cost:

AED 42

Variance Value:

-AED 294

Tolerance Rules

Configure tolerance by:

- Item
- Category
- Location
- Quantity
- Percentage
- Value

Example:

Variance:

1 unit

Value:

AED 3

→ Auto Accept where policy permits.

Variance:

10 units

Value:

AED 4,800

→ Investigation + Approval Required.

Root-Cause Categories

- Counting Error
- Receiving Error
- Transfer Error
- Picking Error
- POS Sync Issue
- F&B Consumption Error
- Damage
- Waste
- Return Error
- Wrong Location
- UOM Error
- Batch Error
- Process Error
- Unknown

Investigation Workspace

Display:

- Expected Stock
- Physical Count
- Variance
- Recent Transactions
- Transfers
- Receipts
- Sales
- Consumption
- Returns
- Adjustments
- Employee / Device History

This allows the controller to investigate before posting an adjustment.

PAGE 7 — Inventory Adjustment & Write-Off Management

Purpose

Provide controlled correction of inventory quantities and values.

Adjustment Types

Support:

- Count Variance
- Damage
- Breakage

- Expiry
- Waste
- Loss
- Found Stock
- Wrong Location Correction
- Batch Correction
- UOM Correction
- Return Adjustment
- System Correction
- Authorized Manual Adjustment

Adjustment Record

Capture:

- Adjustment Number
- Item
- Location
- Batch / Serial
- Current Quantity
- Adjustment Quantity
- New Quantity
- Unit Cost
- Financial Impact
- Reason Code
- Notes
- Supporting Document / Photo
- Created By

Example

ADJ-18422

Item:

Limited Edition Mug

System Quantity:

48

Physical:

45

Adjustment:

-3

Unit Cost:

AED 28

Impact:

-AED 84

Reason:

Breakage confirmed.

Adjustment Lifecycle

Draft

→ Submitted

- Review
- Approval where required
- Posted
- Inventory Ledger Updated
- Finance Entry where applicable

Write-Off

For stock that cannot remain inventory:

Available / Damaged / Expired

↓

Write-Off

↓

Inventory removed from usable stock.

Write-offs shall always retain a reason and audit trail.

PAGE 8 — Inventory Reconciliation & Ledger Control

Purpose

Ensure the Shared Inventory Ledger remains consistent with operational transactions and financial records.

Reconciliation Areas

TICVAI should reconcile:

Inventory Ledger vs Warehouse

Inventory Ledger vs Retail POS

Inventory Ledger vs F&B Consumption

Inventory Ledger vs Procurement Receipts

Inventory Ledger vs Transfers

Inventory Ledger vs Physical Counts

Inventory Ledger vs Finance

Reconciliation Dashboard

Example:

Source	Transactions	Matched	Exceptions	Status
Retail POS	12,480	12,476	4	Warning
F&B	8,220	8,220	0	Reconciled
Procurement	840	838	2	Warning
Transfers	1,240	1,239	1	Warning
Stock Counts	620	620	0	Reconciled

Exception Types

- Missing Inventory Transaction
- Duplicate Transaction
- Quantity Mismatch
- Cost Mismatch
- Location Mismatch
- Batch Mismatch

- Integration Failure
- Posting Failure

Drill-Down

Example:

POS-18428

Retail sold:

2 units

Inventory movement:

1 unit

Variance:

-1

Action:

Review Transaction | Reprocess | Create Adjustment | Escalate

Reconciliation Status

- Matched
- Pending
- Exception
- Investigating
- Resolved
- Approved

PAGE 9 — Shrinkage, Loss & Inventory Accuracy Control

Purpose

Provide management with visibility into inventory accuracy, shrinkage, loss and recurring operational problems.

KPI Cards

- Inventory Accuracy %
- Shrinkage %
- Shrinkage Value
- Damage Value
- Expiry Value
- Waste Value
- Transfer Loss
- Count Variance
- Write-Off Value
- Unexplained Loss

Inventory Accuracy

Track by:

- Company
- Warehouse
- Store
- Outlet
- Category
- Item
- SKU

- Employee / Team where operationally appropriate
- Period

Shrinkage Trend

Example:

January:

0.8%

February:

0.9%

March:

1.4%

April:

1.1%

Target:

<1.0%

Root-Cause Breakdown

Example:

Damage:

32%

Expiry:

24%

Transfer Difference:

14%

Process Error:

12%

Counting Error:

10%

Unknown:

8%

High-Risk Items

Identify inventory with repeated discrepancies.

Example:

Limited Edition Collectible

Count Variance Events:

6

Variance Value:

AED 12,800

Status:

High Risk

Location Benchmark

Compare warehouses/stores by inventory accuracy.

This should be used for operational improvement and controls rather than automatically attributing discrepancies to individual misconduct.

PAGE 10 — AI Inventory Control & Exception Center

Purpose

Provide AI-assisted detection, investigation and recommendation capabilities across transfers, counts, variances and reconciliation.

AI Exception Categories

- Transfer Risk
- Inventory Variance
- Shrinkage Pattern
- Count Anomaly
- Reconciliation Failure
- Repeated Adjustment
- Suspicious Stock Movement Pattern
- Wrong Location
- Batch / Serial Mismatch
- Inventory Accuracy Decline

AI — TRANSFER RECOMMENDATION

TICVAI should identify internal redistribution opportunities before external procurement.

Example:

Main Gift Shop

SKU:

TSH-BLK-M

Available:

8

Forecast Requirement:

52

Shortfall:

44

Waterpark Shop

Available:

86

Forecast Requirement:

32

Excess:

54

Recommendation**Transfer 44 units from Waterpark Shop → Main Gift Shop**

Expected:

Avoided Stock-Out:

44 units

Potential Revenue Protected:

AED 4,180

Confidence:

92%

Actions:

Simulate | Create Transfer | Ignore

AI — TRANSFER RISK

Before approving a transfer, TICVAI should evaluate the impact on the source.

Example:

Requested:

80 units

Source Available:

120

Source Forecast Requirement:

90

Safety Stock:

20

Result:

High Risk

Approving 80 units would create a projected source shortage of 70 units.

Recommendation:

Transfer maximum 10 units or select alternate source.

AI — COUNT PRIORITIZATION

Instead of counting every item equally, TICVAI should recommend count priority based on risk.

Potential factors:

- Item Value
- Sales Velocity
- Historical Variance
- Shrinkage History
- Time Since Last Count
- Manual Adjustments
- Transfer Frequency
- Batch / Serial Sensitivity

Example

SKU-1842

Inventory Value:

AED 84K

Last Count:

42 days ago

Recent Adjustments:

6

Historical Variance:

4.8%

AI Recommendation:

Immediate Cycle Count

AI — VARIANCE ROOT-CAUSE ASSISTANT

Example:

Variance

Item:

Water Bottle

Location:

Main Gift Shop

Variance:

-48 units

AI Analysis

Potential Causes:

1. Transfer Receipt Mismatch — 68% probability

Transfer TR-18420 dispatched 200 but destination receipt recorded 152.

2. POS Sync Issue — 21%

48 sales transactions were delayed during offline period.

3. Counting Error — 11%

Previous count occurred during open trading.

Recommended Investigation**Review TR-18420 first.**

The AI should assist investigation, not automatically post the correction.

AI — SHRINKAGE PATTERN DETECTION

Example:

Inventory variance for souvenir mugs has increased across three locations during the last four weeks.

Observed:

Expected Shrinkage:

0.8%

Actual:

2.7%

Estimated Impact:

AED 18,400

Possible Factors:

- Breakage not consistently recorded
- Transfer receiving differences
- Stock-count process
- POS / inventory synchronization

Recommended Action:

Perform targeted cycle count and review movement history.

AI — RECONCILIATION ASSISTANT

Example:

14 Retail POS transactions have no corresponding inventory movement.

Potential Impact:

27 units

AED 2,840 inventory value

Actions:

Review Transactions | Retry Integration | Create Investigation

The system should never silently create inventory adjustments simply to force reconciliation.

TRANSFER REQUEST FROM OPERATIONAL MODULES

Transfers may originate directly from other TICVAI modules.

Retail Example

Retail Board 4 detects low stock.



Request Internal Transfer



Board 3 checks:

- Source Availability
- Destination Requirement
- Safety Stock
- Existing Transfers
- Forecast Demand



Transfer Created.

F&B Example

Main Restaurant requires:

Beef Patty:

100

Central F&B Warehouse:

Available 420



Transfer Request



Board 3



Board 2 executes:

Pick → Pack → Dispatch



Restaurant Receives



Board 3 reconciles transfer.

This separation is important:

Board 3 controls the transfer transaction.

Board 2 executes warehouse fulfilment.

PROCUREMENT CONNECTION

Before raising external procurement, TICVAI should be capable of checking whether stock exists elsewhere internally.

Example:

Main Gift Shop shortage:

50 units

Central Warehouse:

0

Waterpark Shop:

40 excess

Adventure Store:

20 excess

TICVAI recommendation:

Fulfil internally before purchasing externally.

This gives us the future flow:

Demand

- Check Local Stock
- Check Shared Inventory
- Check Internal Transfer Opportunity
- If insufficient
- **Procurement**

This will become especially important in Boards 4–6.

STOCK FREEZE & COUNT CONTROL

For physical counts, TICVAI should support configurable count policies.

Full Freeze

No inventory movements permitted during count.

Soft Freeze

Transactions continue but are separately tracked and reconciled.

Snapshot Count

System records inventory state at a specific timestamp.

Transactions after that timestamp do not alter the count baseline.

I strongly recommend supporting **Snapshot Count**, because many TICVAI customers—parks, attractions, restaurants and retail stores—cannot stop operations simply to perform inventory counting.

RECOUNT RULES

TICVAI should support automatic recount requirements.

Example:

First Count

System:

100

Physical:

92

Variance:

-8%

Configured Tolerance:

2%

Result:

Recount Required

The second counter may be different from the first where policy requires segregation of duties.

COUNT APPROVAL RULES

Example:

Variance ≤ AED 50

Inventory Supervisor

AED 50–1,000

Inventory Manager

AED 1,000–10,000

Operations / Finance

> AED 10,000

Senior Approval

These thresholds shall be tenant-configurable and use the common TICVAI Approval Engine.

SERIALIZED INVENTORY RECONCILIATION

Serialized items require item-level control.

Example:

System expects:

SER-1001

SER-1002

SER-1003

SER-1004

Physical scan finds:

SER-1001

SER-1002

SER-1004

SER-1007

TICVAI identifies:

Missing: SER-1003

Unexpected: SER-1007

It should not simply show:

System:

4

Physical:

4

Variance:

0.

Quantity matches, but inventory identity does not.

BATCH RECONCILIATION

The same principle applies to batch-controlled inventory.

Example:

Total system quantity:

100

Total physical:

100

But:

Batch A system:

60

Physical:

40

Batch B system:

40

Physical:

60

Result:

Batch-Level Variance

even though total item quantity matches.

This is particularly important for F&B expiry and traceability.

UOM RECONCILIATION

Board 3 shall use Board 1 UOM conversions.

Example:

System:

4 Cases

Each Case:

12 Bottles

Expected:

48 Bottles

Physical:

46 Bottles

Variance:

-2 Bottles

The system should calculate the variance using the correct base UOM and valuation.

MOBILE / EMPLOYEE APP

Inventory employees should be able to execute relevant Board 3 workflows from the TICVAI Employee App.

Transfer

- View Transfer
- Pick / Scan
- Dispatch
- Receive
- Report Difference

Stock Count

- View Assigned Count
- Scan Location
- Scan Item
- Enter Quantity
- Scan Serial / Batch
- Submit Count

Variance

Authorized supervisors may:

- Review Variance
- Add Notes
- Attach Photo
- Request Recount

Adjustment

Authorized roles may:

- Submit Adjustment
 - Select Reason
 - Attach Evidence
-

OFFLINE COUNTING

This is particularly useful for warehouses or venues with unreliable connectivity.

The Employee App should support controlled offline count execution.

Downloaded:

Count Scope + Item References + Locations

↓

Employee performs count offline.



Transactions are time-stamped locally.



Connection restored.



Synchronization.



Conflict detection.

The system shall not silently overwrite inventory changes that occurred during the offline period.

FINANCE INTEGRATION

Inventory adjustments with financial impact shall integrate with Finance.

Example:

Physical Loss:

10 units

Unit Cost:

AED 80

Inventory Adjustment:

-AED 800



Inventory Ledger



Finance Posting / Accounting Interface

Board 3 should provide the inventory transaction and reason.

The Finance module should own the accounting treatment.

APPROVAL ENGINE INTEGRATION

Board 3 shall use TICVAI's shared Approval Engine.

Potential approval events:

- High-Value Transfer
- Emergency Transfer
- Transfer Below Safety Stock
- Large Count Variance
- Manual Adjustment
- Write-Off
- Reconciliation Correction
- Reopening Posted Count

Approval rules may consider:

- Amount
 - Quantity
 - Category
 - Location
 - Reason
 - User
 - Business Unit
-

SEGREGATION OF DUTIES

Board 3 should support separation of responsibilities.

Example:

Counter

should not necessarily be the same user as:

Variance Approver

and:

Adjustment Approver

For sensitive inventory:

Count → Review → Approve → Post

may require separate authorized users.

This shall integrate with TICVAI's common RBAC/PBAC and SoD framework.

AUDIT & TRACEABILITY

Every transfer, count and adjustment shall maintain complete history.

Capture:

- Transaction ID
- Item
- Batch / Serial
- Quantity
- Location
- User
- Device
- Date / Time
- Reason
- Approval
- Previous Quantity
- New Quantity
- Financial Impact
- Source Reference

Example Audit

ADJ-18422

Previous:

48

New:

45

Reason:

Confirmed Breakage

Created:

Inventory Controller

Approved:

Inventory Manager

Financial Impact:

-AED 84

Timestamp:

14 Aug 2026 — 14:42

BOARD 3 — FINAL 10 SCREENS

#	Screen
1	Inventory Control Command Center
2	Inventory Transfer Management
3	Transfer Dispatch, Receipt & Discrepancy Control
4	Stock Count Planning & Scheduling
5	Physical & Cycle Count Execution
6	Variance Investigation & Root-Cause Management
7	Inventory Adjustment & Write-Off Management
8	Inventory Reconciliation & Ledger Control
9	Shrinkage, Loss & Inventory Accuracy Control
10	AI Inventory Control & Exception Center

COMPLETE BOARD 3 FLOW

TRANSFER

Operational Demand

↓
Check Local Stock
↓
Check Enterprise Stock
↓
Identify Internal Source
↓
Transfer Request
↓
Availability + Safety Stock Validation
↓
Approval where required
↓
Board 2:

Pick → Pack → Dispatch

↓
Board 3:

In Transit

↓
Destination:

Receive + Scan + Verify

↓
Quantity Match?

Yes
→ Complete Transfer

No

→ Transfer Discrepancy

→ Investigation

→ Adjustment / Claim / Resolution



Shared Inventory Ledger Updated

STOCK COUNT**Count Plan**

Select:

Warehouse / Store / Zone / Bin / Category / Item



Assign Employees



Create Snapshot / Freeze Policy



Physical Count



Compare:

System vs Physical**Within Tolerance?****Yes**

→ Approve / Post

Outside Tolerance?

→ Recount

→ Variance Investigation

→ Root-Cause Analysis

→ Approval

→ Inventory Adjustment



Shared Inventory Ledger Updated



Finance Impact where applicable



Audit Trail

CONTINUOUS CONTROL**Transfers + Counts + Adjustments + Operational Transactions**

Reconciliation Engine



Detect:

Missing Transaction / Duplicate / Quantity Difference / Cost Difference / Location Difference / Batch Difference



AI Root-Cause Analysis



Investigation



Resolution



Inventory Accuracy Measurement

**Continuous Improvement****IMPORTANT ARCHITECTURAL REQUIREMENT**

Board 3 should not create another inventory balance.

There must remain **one Shared Inventory Ledger**.

The clean architecture is:

Board 1 — Inventory Master & Stock Ledger

What do we own?

**Board 2 — Warehouse Execution**

Where is it and how do warehouse employees physically handle it?

**Board 3 — Inventory Control**

How does it move between locations, how do we verify it, and how do we correct discrepancies?

The distinction between **Board 2 and Board 3** is particularly important.

For example:

Main Gift Shop needs 100 T-Shirts from Central Warehouse.

Board 3 creates and controls the **Transfer**.



Board 2 performs:

Pick → Pack → Stage → Dispatch



Board 3 performs:

Destination Receipt → Quantity Verification → Discrepancy → Reconciliation → Transfer Closure

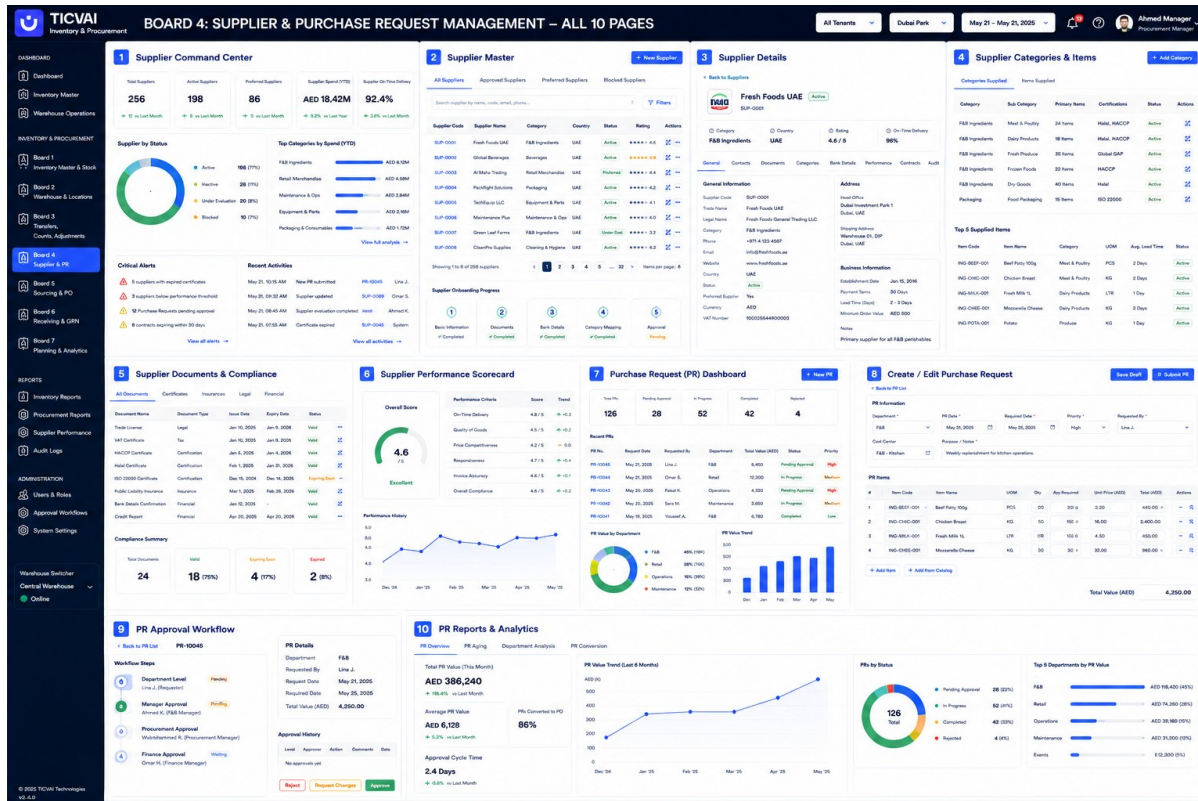
This prevents duplicated transfer logic and gives TICVAI a clean enterprise inventory architecture.

After Board 3, the next major layer should be **Board 4 — Supplier & Procurement Management**, where we move from internal stock control into **Supplier → Purchase Request → Requisition → RFQ → Quotations →**

Comparison → Approval → Purchase Order.

TICVAI INVENTORY & PROCUREMENT — BOARD 4

Supplier, Purchase Request & Procurement Management — Scope of Work



1. Purpose

The Supplier, Purchase Request & Procurement Management Board shall provide TICVAI with the centralized procurement initiation and supplier-management layer for purchasing inventory, consumables, materials, operational supplies, equipment and other approved goods required across the platform.

Boards 1–3 established:

Board 1: What inventory exists and how much stock TICVAI has.

Board 2: Where inventory is physically stored and how warehouse operations execute it.

Board 3: How inventory moves between locations and how stock accuracy is controlled.

Board 4 shall establish:

What do we need to purchase? → Why do we need it? → Can we fulfil it internally first? → Which suppliers can provide it? → Who approves the requirement? → When should procurement begin?

The procurement architecture shall follow:

Demand → Internal Stock Check → Replenishment Requirement → Purchase Request → Approval → Supplier Identification → Procurement Process

This Board shall be used by:

- Inventory
- Warehouse
- Food & Beverage
- Retail
- Rental
- Resource Management
- Maintenance

- Operations
- Events
- Finance
- Department Managers
- Employee App
- AI Replenishment Engine

The key principle is that **Procurement should begin only after TICVAI understands existing enterprise inventory and internal transfer opportunities.**

PAGE 1 — Procurement Command Center

Purpose

Provide Procurement, Inventory, Finance and management with a real-time view of purchasing demand, purchase requests, supplier activity, approvals and procurement risks.

Global Filters

- Tenant
- Company
- Business Unit
- Park
- Venue
- Department
- Warehouse
- Category
- Supplier
- Request Status
- Priority
- Buyer
- Date / Period

KPI Cards

- Open Purchase Requests
- PR Value
- Pending Approval
- Approved Requests
- Urgent Requests
- Auto-Generated Requests
- Requests Awaiting Procurement
- Active Suppliers
- Suppliers with Issues
- Procurement Lead Time
- Potential Internal Fulfilment
- Budget Exceptions
- Critical Stock Risks

Procurement Pipeline

Display:

Demand Identified

→ **Internal Stock Check**

→ **PR Draft**

→ **Submitted**

→ **Approval**

→ **Approved**

→ **Ready for RFQ / PO**

Example:

Demand Identified: **42**

Draft PR: **18**

Pending Approval: **14**

Approved: **26**

Ready for RFQ: **11**

Urgent: **5**

Procurement by Business Area

Business Area	Open PRs	Value	Urgent	Approval	Status
F&B	18	AED 184K	4	6	Attention
Retail	12	AED 126K	2	3	Healthy
Operations	8	AED 92K	1	2	Healthy
Maintenance	7	AED 148K	2	4	Attention

Procurement Alerts

Examples:

14 inventory items are projected to fall below safety stock before supplier lead time.

PR-18422 has been awaiting Finance approval for 18 hours.

Water Bottles requested by Main Gift Shop are available at two other locations.

Supplier SUP-0042 has three recent late deliveries.

Quick Actions

+ [Purchase Request](#) | [Supplier Search](#) | [Internal Stock Check](#) | [Approval Center](#) | [View Procurement Pipeline](#)

PAGE 2 — Supplier Master & Supplier Profile

Purpose

Maintain centralized supplier records that may be used across all TICVAI procurement activities.

There should be **one Supplier Master**, not separate supplier databases for Retail, F&B and other modules.

Supplier General Information

Capture:

- Supplier Name
- Supplier Code
- Legal Name
- Supplier Type
- Registration Number
- Tax / VAT Number

- Country
- Address
- Contact Person
- Email
- Phone
- Website
- Currency
- Language
- Status

Supplier Types

Examples:

- F&B Supplier
- Retail Merchandise Supplier
- General Supplier
- Equipment Supplier
- Maintenance Supplier
- Packaging Supplier
- Service Supplier
- Manufacturer
- Distributor

A supplier may belong to multiple types.

Commercial Information

Capture:

- Payment Terms
- Default Currency
- Minimum Order Value
- Minimum Order Quantity
- Standard Lead Time
- Delivery Terms
- Incoterms where applicable
- Freight Terms
- Contract Reference
- Contract Start
- Contract End

Supplier Categories

A supplier may be approved for specific:

- Categories
- Subcategories
- Brands
- Items
- Locations
- Companies

Supplier Status

- Draft
- Under Review

- Approved
- Active
- Suspended
- Blocked
- Expired
- Archived

Supplier Documents

Support attachments such as:

- Trade License
- Tax Certificate
- Bank Information
- Insurance
- Certifications
- Food Safety Documents
- Contracts
- Product Certifications
- Other Compliance Documents

PAGE 3 — Supplier Qualification, Compliance & Performance

Purpose

Control whether suppliers are permitted to participate in procurement and continuously evaluate their performance.

Qualification Criteria

Configurable criteria may include:

- Commercial Registration
- Tax Registration
- Bank Verification
- Insurance
- Quality Certification
- Food Safety Certification
- Product Certification
- Sustainability Requirement
- Contract Status
- Approved Product Categories

Compliance Status

Display:

Valid

Expiring Soon

Expired

Missing

Example

Fresh Foods UAE LLC

Trade License:

Valid — 184 days

Food Safety Certificate:

Valid — 92 days

Insurance:

Expires in 14 days

Status:

Warning

Supplier Performance KPIs

- On-Time Delivery %
- Order Fill Rate
- Quality Acceptance %
- Quantity Accuracy
- Price Variance
- Return Rate
- Damage Rate
- Lead-Time Accuracy
- Invoice Accuracy
- Response Time
- Dispute Rate

Supplier Score

Example:

SUP-0042 — Fresh Foods UAE

Quality:

94%

Delivery:

82%

Price:

89%

Compliance:

100%

Overall:

89 / 100

Performance Classification

- Preferred
- Approved
- Conditional
- Watchlist
- Suspended

Automatic Alerts

Example:

Supplier delivery performance has fallen below 85% for three consecutive months.

The system may recommend review but should not automatically block the supplier unless configured policy allows it.

PAGE 4 — Supplier Item Catalog & Commercial Agreements

Purpose

Connect suppliers with the Inventory Item Master from Board 1 and maintain supplier-specific purchasing information.

Supplier Item Record

For each Supplier + Inventory Item:

- Supplier Item Code
- TICVAI Inventory Item
- Supplier Description
- Purchase UOM
- Pack Size
- Minimum Order Quantity
- Unit Price
- Currency
- Lead Time
- Tax
- Contract Price
- Effective Date
- Expiry Date
- Preferred Supplier
- Alternative Supplier
- Status

Example

Inventory Item

Water Bottle 500ml

Supplier A

Price:

AED 1.85

MOQ:

1,000

Lead Time:

3 Days

Supplier B

Price:

AED 1.72

MOQ:

5,000

Lead Time:

7 Days

Supplier C

Price:

AED 2.05

MOQ:

500

Lead Time:

1 Day

TICVAI therefore should not determine "best supplier" based only on lowest unit price.

Commercial Agreements

Support:

- Contract Price
- Volume Price
- Tiered Price
- Promotional Price
- Framework Agreement
- Fixed Period Price
- Quantity Break

Price History

Maintain historical supplier pricing.

Example:

Effective	Supplier	Item	Price
Jan	Supplier A	Water Bottle	AED 1.70
Apr	Supplier A	Water Bottle	AED 1.78
Aug	Supplier A	Water Bottle	AED 1.85

PAGE 5 — Purchase Request Creation & Management

Purpose

Allow departments and operational modules to request the purchase of goods.

Purchase Request Sources

A PR may originate from:

- Manual Employee Request
- Inventory Reorder
- Low Stock
- Forecast Requirement
- F&B Requirement
- Retail Requirement
- Event Requirement
- Maintenance Requirement
- Resource Requirement
- Warehouse Requirement
- AI Recommendation

PR Header

Capture:

- PR Number
- Company
- Business Unit
- Department
- Venue
- Requester
- Required Date

- Priority
- Reason
- Budget / Cost Center
- Project / Event
- Delivery Location

PR Lines

Capture:

- Inventory Item
- Description
- Quantity
- UOM
- Estimated Unit Cost
- Estimated Total
- Required Date
- Suggested Supplier
- Notes

Non-Catalog Request

Where policy permits, users may request an item that does not yet exist in the Item Master.

Example:

New specialized maintenance component

This should trigger appropriate master-data review before procurement completion.

Supporting Documents

Allow:

- Specifications
- Images
- Supplier Proposal
- Technical Document
- Justification
- Event Requirement

PAGE 6 — Internal Stock & Transfer Check Before Procurement

Purpose

Prevent unnecessary purchasing when suitable inventory already exists somewhere inside the TICVAI enterprise.

This is a particularly important TICVAI feature.

Before a PR proceeds to external procurement, the system should check:

- Local Available Stock
- Warehouse Stock
- Other Venue Stock
- Other Store Stock
- Other Outlet Stock
- In-Transit Stock
- On-Order Stock
- Reserved Stock
- Safety Stock

- Forecast Demand

Example

Main Gift Shop requests:

100 Water Bottles

Local Available:

12

Central Warehouse:

0

Waterpark Shop Excess:

58

Adventure Store Excess:

42

TICVAI Recommendation**Internal Transfer Available**

Transfer:

58 from Waterpark

42 from Adventure Store

External Purchase Required:

0

Actions:

Create Transfers | Continue Procurement | Modify Request

Partial Internal Fulfilment

Example:

Required:

500

Internal Excess:

180

Remaining:

320

TICVAI recommends:

Transfer 180 internally + Procure 320 externally

Override

If user continues procurement despite available internal stock:

Reason Required

Examples:

- Different Batch Required
 - Event-Dedicated Stock
 - Internal Stock Not Suitable
 - Transfer Lead Time Too Long
 - Management Decision
-

PAGE 7 — Auto-Replenishment & System-Generated Purchase Requests

Purpose

Allow TICVAI to automatically create procurement demand based on inventory policies and operational forecasts.

Trigger Sources

- Minimum Stock
- Reorder Point
- Safety Stock
- Par Level
- Forecast Demand
- Event Demand
- Seasonal Requirement
- Production Requirement
- Supplier Lead Time
- Existing On-Order Quantity

Example

Beef Patty

Available:

420

Safety Stock:

200

Forecast 5-Day Consumption:

580

Incoming:

100

Projected Position:

-60

Recommended Purchase:

260 units

System Action

Depending on tenant policy:

Option A

Generate Recommendation Only

Option B

Create Draft PR

Option C

Create PR and Submit for Approval

I recommend **Option B — Create Draft PR** as the standard TICVAI behavior.

This preserves human control while removing repetitive work.

Auto-PR Information

The generated request should explain:

Why it was created.

Example:

Created because projected inventory will fall below safety stock on 18 Aug based on current demand and supplier lead time.

PAGE 8 — Purchase Request Approval & Budget Control

Purpose

Apply configurable approval workflows before procurement commits to external sourcing.

Approval Inputs

Rules may consider:

- PR Value
- Department
- Item Category
- Company
- Cost Center
- Budget
- Requester
- Supplier
- Priority
- Emergency Status

Example Approval Matrix

Up to AED 5,000

Department Manager

AED 5,001–25,000

Department Manager + Finance

AED 25,001–100,000

Director + Finance

Above AED 100,000

Senior Management

These thresholds must be tenant-configurable.

Budget Validation

Where Finance provides budget information:

Requested:

AED 84,000

Available Budget:

AED 112,000

Remaining After PR:

AED 28,000

Result:

Within Budget

Budget Exception

Requested:

AED 84,000

Available:

AED 60,000

Result:

Budget Exception

Actions:

Return | Modify | Request Exception Approval

Approval Actions

- Approve
- Reject
- Return for Modification
- Delegate
- Request Information

No separate procurement approval engine shall be built; Board 4 shall use TICVAI's common Approval Engine.

PAGE 9 — Procurement Demand Consolidation & Sourcing Preparation

Purpose

Allow Procurement teams to consolidate approved requirements before initiating RFQ or Purchase Order activity.

Example

Approved requirements:

Main Restaurant:

100 units

VIP Lounge:

60 units

Beach Kiosk:

40 units

Retail Event:

100 units

Same Inventory Item:

Total Requirement = 300 units

Instead of issuing multiple supplier requests, Procurement may consolidate them.

Consolidation Criteria

Support grouping by:

- Inventory Item
- Category
- Supplier
- Required Date
- Delivery Location
- Company
- Currency
- Contract

Procurement Workbench

Display:

Requirement	Item	Qty	Needed	Supplier Options	Action
PR-18421	Beef Patty	300	18 Aug	3	RFQ
PR-18422	Water Bottle	2,000	20 Aug	4	RFQ

Requirement	Item	Qty	Needed	Supplier Options	Action
PR-18423	T-Shirt	500	28 Aug	2	Direct PO

Sourcing Route

Authorized users select:

RFQ

or

Direct Purchase Order

or

Existing Contract

or

Framework Agreement

according to procurement policy.

Direct PO Rules

Example:

Existing contracted supplier + agreed price + value below threshold

→ Direct PO permitted.

Otherwise:

→ RFQ required.

The RFQ, quotation comparison and PO process will be expanded in Board 5.

PAGE 10 — Procurement AI, Exceptions & Action Center

Purpose

Provide centralized intelligence for procurement demand, supplier risk, stock availability, budget risk and purchasing opportunities.

Exception Categories

- Urgent Stock Requirement
- PR Approval Delay
- Duplicate PR
- Internal Stock Available
- Budget Exception
- Supplier Compliance Risk
- Supplier Lead-Time Risk
- Price Increase
- Contract Expiring
- Repeated Emergency Purchase
- Excessive Manual Procurement

AI — PURCHASE NEED RECOMMENDATION

Example:

Item

Water Bottle 500ml

Available:

1,420

In Transit:

500

Forecast 14-Day Demand:

3,800

Safety Stock:

600

Expected Position:

-1,480

Recommendation

Procure:

1,500 units

Required Delivery:

Before 22 Aug

Confidence:

94%

Explanation

Based on:

- Current Inventory
- Existing Transfers
- Existing POs
- Sales Velocity
- Ticket Forecast
- Event Schedule
- Safety Stock

AI — INTERNAL FULFILMENT BEFORE PURCHASE

Example:

PR-18428 requests 120 Logo Caps.

TICVAI finds:

Central Warehouse:

0

Waterpark:

80 excess

Event Warehouse:

50 excess

Recommendation:

Transfer 120 internally instead of purchasing.

Estimated Purchase Avoidance:

AED 3,240

Actions:

Create Transfer | Continue PR | Review Stock

AI — DUPLICATE DEMAND DETECTION

Example:

F&B Operations submits:

PR-18420:

500 Water Bottles

Retail submits:

PR-18426:

800 Water Bottles

Event Operations submits:

PR-18431:

1,200 Water Bottles

All required within:

3 days.

AI Recommendation:

Consolidate into one procurement requirement for 2,500 units to improve sourcing efficiency and potential volume pricing.

AI — SUPPLIER RECOMMENDATION

TICVAI should evaluate suppliers using more than price.

Example:

Factor	Supplier A	Supplier B	Supplier C
Price	AED 1.85	AED 1.72	AED 2.05
Lead Time	3d	7d	1d
Delivery Score	96%	84%	98%
Quality	94%	96%	97%
MOQ	1,000	5,000	500

Required:

1,500 units

Needed:

In 3 days

AI Recommendation

Supplier A

Reason:

- Meets required date
- Strong delivery history
- Acceptable MOQ
- Lower total risk than Supplier B
- Lower cost than Supplier C

Confidence:

91%

This is a recommendation only; supplier selection remains governed by procurement policy and approvals.

AI — PRICE ANOMALY DETECTION

Example:

Supplier quotation:

AED 2.15/unit

Last Purchase:

AED 1.82

Contract Reference:

AED 1.85

Increase:

+16.2%

TICVAI Alert:

Proposed supplier price materially exceeds recent and contracted pricing.

Actions:

Review History | Request Clarification | Compare Suppliers

AI — PROCUREMENT TIMING

TICVAI should recommend **when** to purchase, not only how much.

Example:

Current Stock:

2,400

Forecast Daily Demand:

180

Safety Stock:

600

Supplier Lead Time:

5 Days

Recommended Order Date:

18 Aug

Recommended Delivery:

23 Aug

This reduces unnecessary early purchasing and inventory holding.

AI — EMERGENCY PROCUREMENT ANALYSIS

TICVAI should monitor repeated urgent purchases.

Example:

F&B has raised 14 emergency PRs for Fresh Milk during the past 60 days.

Potential Cause:

Reorder Point Too Low

or

Supplier Lead-Time Configuration Inaccurate

Recommendation:

Review replenishment policy instead of continuing emergency purchasing.

This connects procurement intelligence back to Board 1 stock policies and Board 6 forecasting.

PURCHASE REQUEST LIFECYCLE

The standard lifecycle should be:

Draft

→ Internal Stock Check

→ Submitted

→ Pending Approval

→ Approved

→ Procurement Review

→ Consolidated where applicable

→ RFQ / Contract / Direct PO

→ Converted

→ Closed

Alternative statuses:

- Returned
- Rejected
- Cancelled
- Partially Converted
- On Hold

SUPPLIER ONBOARDING WORKFLOW

Supplier onboarding should follow:

Supplier Request

↓

Supplier Data

↓

Required Documents

↓

Compliance Review

↓

Commercial Review

↓

Finance / Bank Validation where applicable

↓

Approval

↓

Active Supplier

Expiry Monitoring

TICVAI should monitor supplier documentation.

Example:

Supplier food-safety certification expires in 14 days.

Depending on policy:

Warning

or

Prevent New Orders after Expiry

SUPPLIER DUPLICATE PREVENTION

Before creating a supplier, TICVAI should search:

- Legal Name
- Registration Number
- Tax Number
- Bank Details where permitted
- Email
- Phone

Example:

Potential duplicate supplier detected: Fresh Foods UAE LLC / Fresh Food UAE L.L.C.

Actions:

Compare | Merge Request | Continue Creation

PROCUREMENT CATEGORY MANAGEMENT

Procurement should support category ownership.

Examples:

F&B

- Fresh Food
- Frozen
- Dry Goods
- Beverages
- Packaging

Retail

- Apparel
- Souvenirs
- Toys
- Merchandise

Operations

- Cleaning
- Consumables
- Maintenance
- Equipment

Each category may have:

- Category Manager
- Approved Suppliers
- Procurement Policy
- RFQ Threshold
- Approval Rules
- Contract Requirements

CONTRACT / AGREEMENT REFERENCE

Board 4 does not need to become a complete Contract Management System, but procurement should be able to reference supplier commercial agreements.

Capture:

- Agreement Number
- Supplier
- Effective Date
- Expiry
- Categories
- Items
- Agreed Prices
- Spend Limit
- Document

Contract Alert

Supplier contract SUP-C-1842 expires in 30 days.

F&B INTEGRATION

Example:

F&B Board identifies:

Beef Patty projected shortage.



Shared Inventory checks:

Main Restaurant

Other Outlets

F&B Warehouse

Central Warehouse

In Transit

On Order



Internal supply insufficient.



Board 4 creates:

Draft Purchase Request



Required:

500 units

Required Date:

18 Aug

Reason:

Forecast Consumption + Safety Stock



Approval



Procurement.

RETAIL INTEGRATION

Example:

Retail forecast identifies:

Concert T-Shirt requirement:

2,150

Available + Incoming:

1,820

Shortfall:

330

Safety Stock:

100

Recommended Purchase:

450 units

Board 4 creates:

Draft PR — Event Merchandise

with event reference automatically attached.

MAINTENANCE / RESOURCE INTEGRATION

Example:

Resource Management identifies maintenance requirement:

Replacement Scanner Batteries:

100

Available:

20

Required:

120



Procurement Requirement:

100

The same procurement engine is used.

We should not create separate purchasing systems for F&B, Retail, Maintenance or Resources.

EMPLOYEE APP INTEGRATION

Authorized employees should be able to:

- Create Purchase Request
- Search Inventory Item
- Enter Quantity
- Select Required Date
- Add Justification
- Attach Photo / Document
- View PR Status
- Respond to Returned Request
- Approve / Reject where authorized

Example

Purchase Request PR-18422

Status:

Pending Finance Approval

Required:

20 Aug

Estimated:

AED 18,400

NOTIFICATIONS

Notifications may be generated for:

- PR Submitted
- Approval Required
- PR Approved
- PR Rejected
- PR Returned
- Supplier Document Expiring
- Budget Exception
- Urgent Requirement
- Procurement Delay
- Supplier Risk

Channels may include:

- Employee App
- In-App Notification

- Email
- Other configured enterprise channels

APPROVAL ENGINE

Board 4 shall consume the common TICVAI Approval Engine.

Approval workflows should support:

- Sequential Approval
- Parallel Approval
- Conditional Approval
- Value-Based Approval
- Category-Based Approval
- Department-Based Approval
- Emergency Approval
- Delegated Approval

Example

PR = AED 84,000

Department Manager

↓

Procurement Manager

↓

Finance

↓

Director

↓

Approved

The procurement module should display approval status but should not maintain a separate approval framework.

SEGREGATION OF DUTIES

Procurement should support controls such as:

Requester ≠ Final Approver

Supplier Creator ≠ Supplier Approver

Buyer ≠ Final High-Value PO Approver

where tenant policy requires it.

These controls shall use TICVAI's common RBAC/PBAC and SoD architecture.

AUDIT & TRACEABILITY

Board 4 shall retain complete history for:

Supplier

- Creation
- Changes
- Compliance
- Status
- Approvals

Purchase Request

- Request
- Modification
- Approval

- Rejection
- Internal Stock Recommendation
- Conversion

Commercial Data

- Supplier Price
- MOQ
- Lead Time
- Contract Reference

Audit record:

User + Date/Time + Action + Previous Value + New Value + Reason + Approval

BOARD 4 — FINAL 10 SCREENS

#	Screen
1	Procurement Command Center
2	Supplier Master & Supplier Profile
3	Supplier Qualification, Compliance & Performance
4	Supplier Item Catalog & Commercial Agreements
5	Purchase Request Creation & Management
6	Internal Stock & Transfer Check Before Procurement
7	Auto-Replenishment & System-Generated Purchase Requests
8	Purchase Request Approval & Budget Control
9	Procurement Demand Consolidation & Sourcing Preparation
10	Procurement AI, Exceptions & Action Center

COMPLETE BOARD 4 FLOW

OPERATIONAL DEMAND

Retail / F&B / Warehouse / Event / Maintenance / Resource Management

↓

Inventory Requirement

↓

SHARED INVENTORY CHECK

Local Available

↓

Warehouse Available

↓

Other Locations

↓

In Transit

↓

On Order

↓

Safety Stock / Forecast



CAN REQUIREMENT BE FULFILLED INTERNALLY?

YES

→ Board 3 Internal Transfer

→ Board 2 Warehouse Execution

→ Requirement Fulfilled

PARTIALLY

→ Internal Transfer

•
Purchase Remaining Quantity

NO

→ Purchase Request



Budget / Cost Center Validation



Approval



Procurement Consolidation



Supplier / Contract Evaluation



RFQ / Direct PO



Board 5

PROCUREMENT INTELLIGENCE LOOP

Current Stock



Consumption / Sales



Forecast Demand



Ticket / Event Forecast



Safety Stock



Existing Transfers



Open Purchase Orders



Supplier Lead Time



TICVAI PROCUREMENT INTELLIGENCE



Recommend:

Transfer Internally

or

Wait

or

Purchase



If Purchase:

What Quantity?

When?

Which Supplier Options?

What Expected Cost?

↓

Human Review / Approval

↓

Procurement Execution

↓

Supplier Performance Measurement

↓

Future Recommendation Improvement

IMPORTANT ARCHITECTURAL REQUIREMENT

I recommend we make one boundary very clear to SoftLab:

Board 4 owns:

Supplier Master + Supplier Qualification + Supplier Catalog + Purchase Request + Internal Stock Check + Approval + Demand Consolidation

It should **not yet own the complete RFQ, quotation and Purchase Order transaction lifecycle.**

That belongs to the next board.

Therefore:

Board 4 — Procurement Initiation

Demand → PR → Approval → Sourcing Requirement

↓

Board 5 — Sourcing & Purchase Order

RFQ → Supplier Quotes → Comparison → Negotiation → Award → PO

↓

Later:

Receiving / GRN / Invoice Matching

This separation is important because otherwise Board 4 becomes overloaded and SoftLab may mix **requesting something** with **commercially purchasing it**.

The other architectural point I strongly recommend is the **Internal Fulfilment Gate**:

Before TICVAI purchases externally, it should always understand whether the requirement can be fulfilled from existing enterprise inventory.

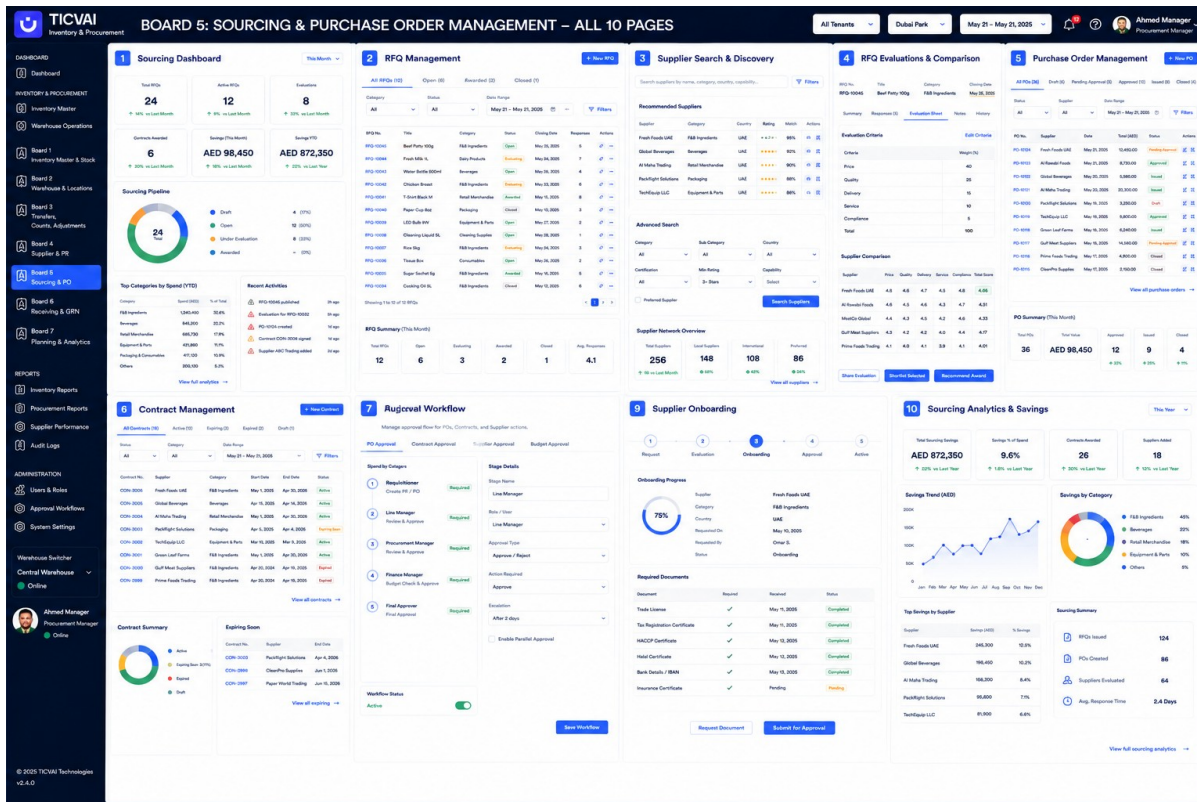
That gives TICVAI a much stronger workflow than a conventional procurement system:

Need → Check → Transfer → Procure only the remaining requirement.

It connects the work we already completed across **F&B, Retail, Shared Inventory, Warehouse and Inventory Control** into one coherent procurement architecture.

TICVAI INVENTORY & PROCUREMENT — BOARD 5

RFQ, Supplier Quotation, Sourcing & Purchase Order Management — Scope of Work



1. Purpose

The RFQ, Supplier Quotation, Sourcing & Purchase Order Management Board shall provide TICVAI with the centralized commercial procurement execution layer for converting approved purchasing demand into controlled supplier sourcing and Purchase Orders.

Board 4 established:

What needs to be purchased → Internal stock check → Purchase Request → Approval → Consolidated procurement demand.

Board 5 shall establish:

How do we source it? → Which suppliers should be invited? → What did each supplier offer? → Which offer provides the best overall value? → Who approves the award? → How is the Purchase Order created and controlled?

The core lifecycle shall follow:

Approved Demand → Sourcing Strategy → RFQ → Supplier Quotations → Commercial / Technical Evaluation → Comparison → Negotiation → Award Recommendation → Approval → Purchase Order → Supplier Acknowledgement → Order Tracking

The Board shall support purchasing for:

- Food & Beverage
- Retail
- Inventory Replenishment
- Operations
- Maintenance
- Rental

- Equipment
- Events
- Packaging
- Consumables
- General Procurement

The Board shall integrate with TICVAI's shared:

- Inventory Engine
- Supplier Master
- Approval Engine
- Finance
- Budget Control
- Warehouse
- Receiving / GRN
- Employee App
- Analytics & AI

PAGE 1 — Sourcing & Purchase Order Command Center

Purpose

Provide Procurement management with a real-time overview of RFQs, supplier responses, sourcing decisions, Purchase Orders and commercial exceptions.

Global Filters

- Tenant
- Company
- Business Unit
- Procurement Category
- Buyer
- Supplier
- RFQ Status
- PO Status
- Priority
- Required Date
- Date / Period

KPI Cards

- Open RFQs
- RFQ Value
- Suppliers Invited
- Quotations Received
- Quotations Pending
- RFQs Closing Today
- Awards Pending Approval
- Purchase Orders Created
- Open PO Value
- POs Awaiting Supplier Confirmation
- Late POs

- Savings vs Estimate
- Contract Coverage %
- Critical Sourcing Alerts

Sourcing Pipeline

Display:

Approved Demand

- RFQ Preparation
- RFQ Issued
- Supplier Response
- Evaluation
- Negotiation
- Award
- PO Creation
- Supplier Confirmation

Example:

Approved Demand: 24

RFQ Draft: 8

RFQ Issued: 12

Evaluation: 7

Award Pending: 4

PO Created: 18

Supplier Confirmation Pending: 5

Category Spend Snapshot

Category	Open Sourcing	PO Value	Savings	Risk
F&B	8	AED 620K	AED 42K	Medium
Retail	6	AED 410K	AED 31K	Low
Maintenance	5	AED 380K	AED 18K	High
Operations	3	AED 190K	AED 9K	Low

Alerts

Examples:

- RFQ-18421 closes in 2 hours with only one supplier response.
- Supplier SUP-0042 increased price 16.2% versus last purchase.
- PO-18438 has not been acknowledged by supplier within SLA.
- Award recommendation exceeds approved PR budget by AED 12,800.

Quick Actions

+ RFQ | Compare Quotes | Award Review | + Purchase Order | Supplier Follow-Up | View Exceptions

PAGE 2 — RFQ Creation & Sourcing Strategy

Purpose

Create a formal Request for Quotation from approved procurement demand.

RFQ Sources

An RFQ may originate from:

- Single Approved PR
- Multiple Consolidated PRs
- Replenishment Requirement
- Event Requirement
- Framework Re-Sourcing
- Contract Expiry
- Manual Authorized Sourcing

RFQ Header

Capture:

- RFQ Number
- Company
- Procurement Category
- Buyer
- Requesting Department
- Delivery Location
- Required Delivery Date
- Currency
- Submission Deadline
- Evaluation Method
- Commercial Terms
- Notes

RFQ Lines

For each item:

- Inventory Item
- Description
- Specification
- Quantity
- Purchase UOM
- Required Date
- Estimated Price
- Budget Reference
- Delivery Location

Sourcing Strategy

Configure:

- Open to Approved Suppliers
- Selected Supplier List
- Preferred Suppliers Only
- Contracted Suppliers
- Emergency Sourcing
- Single-Source Procurement
- Competitive RFQ

Minimum Responses

Example:

Standard Procurement:

Minimum **3 Quotations**

Emergency:

Minimum **1**

Strategic / High Value:

Configurable policy.

Documents

Attach:

- Specifications
- Drawings
- Product Images
- Technical Requirements
- Terms & Conditions
- Compliance Requirements

PAGE 3 — Supplier Invitation & RFQ Response Management

Purpose

Manage supplier participation and quotation submission.

Supplier Selection

TICVAI should recommend eligible suppliers based on:

- Approved Category
- Item Capability
- Compliance Status
- Supplier Status
- Delivery Geography
- Historical Performance
- Contract Status

Supplier Invitation

Track:

- Supplier
- Invitation Date
- Contact
- Delivery Status
- Viewed
- Response Intent
- Quotation Submitted
- Declined
- No Response

Supplier Response Status

- Invited
- Viewed
- Participating
- Submitted
- Declined

- Late
- Disqualified

Supplier Portal / Integration

The architecture should support a future supplier-facing portal or secure external submission mechanism allowing suppliers to:

- View RFQ
- Download Specification
- Enter Prices
- Enter Lead Time
- Enter MOQ
- Confirm Tax
- Confirm Delivery
- Upload Quote
- Submit Clarification

Clarification Management

Buyer and supplier should be able to record controlled RFQ clarifications.

Example:

Supplier asks:

Is equivalent packaging acceptable?

Buyer response:

Yes, subject to food-grade certification.

All clarifications should remain part of RFQ history.

PAGE 4 — Supplier Quotation Management

Purpose

Capture and normalize supplier offers for comparison.

Quotation Header

Capture:

- Quotation Number
- Supplier
- RFQ
- Quote Date
- Valid Until
- Currency
- Payment Terms
- Delivery Terms
- Lead Time
- Freight
- Tax
- Warranty
- Notes

Quotation Lines

Capture:

- Item

- Supplier Item Code
- Offered Quantity
- Unit Price
- Discount
- Tax
- Freight Allocation
- Total Cost
- MOQ
- Pack Size
- Lead Time
- Delivery Date
- Substitute Item where permitted

Landed Cost

Where applicable, TICVAI should support comparison using:

Unit Price + Freight + Duty + Applicable Charges = Estimated Landed Cost

rather than comparing only headline unit price.

Alternative Offer

Example:

Requested:

Brand A Water Bottle

Supplier proposes:

Brand B equivalent

The alternative should be clearly flagged and require acceptance where specifications differ.

Quote Validation

TICVAI should identify:

- Missing Price
- Invalid UOM
- Price Currency Mismatch
- Delivery After Required Date
- MOQ Above Requirement
- Expired Quote
- Unapproved Substitute

PAGE 5 — Quotation Comparison & Supplier Evaluation

Purpose

Provide a side-by-side sourcing evaluation rather than manually comparing supplier spreadsheets.

Comparison Example

Criteria	Supplier A	Supplier B	Supplier C
Unit Price	AED 1.85	AED 1.72	AED 2.05
Landed Cost	AED 1.93	AED 1.84	AED 2.08
Lead Time	3 Days	7 Days	1 Day
MOQ	1,000	5,000	500

Criteria	Supplier A	Supplier B	Supplier C
Delivery Score	96%	84%	98%
Quality Score	94%	96%	97%
Compliance	Pass	Pass	Pass
Total Score	91	82	89

Evaluation Dimensions

Support weighted evaluation based on:

- Price
- Total Cost
- Lead Time
- Quality
- Delivery Performance
- MOQ
- Compliance
- Payment Terms
- Warranty
- Supplier Risk
- Sustainability where applicable
- Technical Evaluation

Configurable Weighting

Example:

Price:
35%

Quality:
25%

Delivery:
20%

Lead Time:
10%

Compliance:
10%

Technical vs Commercial Evaluation

For appropriate procurement:

Technical Evaluation

and

Commercial Evaluation

should be separable.

Commercial users should not necessarily influence technical scoring before technical qualification is complete.

Recommendation

TICVAI may recommend:

Best Overall Value

not merely:

Lowest Price

PAGE 6 — Negotiation & Best-and-Final-Offer Management

Purpose

Support controlled supplier negotiation and revised quotations.

Negotiation Round

Example:

RFQ-18421

Round 1

Supplier A:

AED 184K

Supplier B:

AED 176K

Supplier C:

AED 196K

Buyer initiates:

Best & Final Offer

Supplier Revision

Track:

- Original Quote
- Revised Quote
- Revision Number
- Revision Date
- Price Change
- Term Change
- Delivery Change
- Notes

Negotiation Log

Capture:

- Meeting
- Call
- Email Summary
- Commercial Clarification
- Requested Revision
- Supplier Response

Savings

TICVAI should calculate:

Initial Estimate

vs

Initial Quote

vs

Final Negotiated Price

Example:

Budget Estimate:

AED 200K

Initial Best Quote:

AED 186K

Final Award:

AED 174K

Savings vs Estimate:

AED 26K / 13%

Governance

Negotiation history should not be overwritten.

Each commercial revision shall be retained.

PAGE 7 — Award Recommendation & Approval

Purpose

Formally recommend the selected supplier and obtain required approvals before commercial commitment.

Award Recommendation

Capture:

- RFQ
- Recommended Supplier
- Recommended Lines
- Total Award Value
- Delivery Date
- Commercial Terms
- Evaluation Score
- Savings
- Reason for Selection
- Buyer Recommendation

Split Award

TICVAI should support awarding different lines or quantities to different suppliers.

Example:

Supplier A:

60% quantity

Supplier C:

40% quantity

Reason:

Capacity / Delivery Risk.

Award Scenarios

- Single Supplier
- Split Supplier
- Multi-Supplier Framework
- Partial Award
- No Award / Re-Tender

Approval Validation

Before submission:

TICVAI checks:

- ✓ PR Approved
- ✓ Budget Available / Exception Approved
- ✓ Supplier Approved

- ✓ Supplier Compliance Valid
- ✓ Quote Valid
- ✓ Required Evaluation Complete
- ✓ Award within Procurement Policy

Approval Actions

Approve | Reject | Return | Request Clarification

Deviation Approval

If buyer selects a supplier that is not the top system-ranked supplier:

Reason Required

Example:

Supplier A selected over cheaper Supplier B due to delivery date requirement.

This maintains transparency without forcing the AI recommendation.

PAGE 8 — Purchase Order Creation & Management

Purpose

Convert approved sourcing awards or approved direct-purchase requirements into formal Purchase Orders.

PO Creation Sources

- Approved RFQ Award
- Direct PO
- Existing Contract
- Framework Agreement
- Emergency Procurement

PO Header

Capture:

- PO Number
- Supplier
- Company
- Buyer
- Currency
- Order Date
- Delivery Location
- Required Date
- Payment Terms
- Delivery Terms
- Contract Reference
- Cost Center
- Budget
- Status

PO Lines

- Inventory Item
- Description
- Supplier Item
- Quantity
- UOM

- Unit Price
- Discount
- Tax
- Freight
- Total
- Delivery Date

PO Totals

Display:

Subtotal

Discount

Freight

Tax

Other Charges

Total PO Value**PO Lifecycle****Draft**

- Approval if required
- Issued
- Supplier Acknowledged
- Partially Delivered
- Fully Delivered
- Closed

Alternative statuses:

- On Hold
- Amended
- Cancelled
- Disputed

PAGE 9 — PO Amendment, Supplier Acknowledgement & Order Tracking

Purpose

Control changes after a PO has been issued and track supplier commitment.

Supplier Acknowledgement

Supplier should confirm:

- PO Received
- Quantity Accepted
- Price Accepted
- Delivery Date
- Terms
- Exceptions

Acknowledgement Status

- Awaiting
- Confirmed
- Confirmed with Changes

- Rejected
- Overdue

PO Amendment

Changes may include:

- Quantity
- Price
- Delivery Date
- Delivery Location
- Item
- Commercial Terms

Amendment Control

Example:

PO Value:

AED 184,000

Proposed Change:

+AED 28,000

Change:

+15.2%

Result:

Reapproval Required

Versioning

PO V1

↓

PO V2 Amendment

↓

PO V3 Final

All versions must remain traceable.

Delivery Schedule

Support multiple scheduled deliveries.

Example:

PO 1,000 units

Delivery 1:

400 — 20 Aug

Delivery 2:

300 — 27 Aug

Delivery 3:

300 — 3 Sep

Order Tracking

Display:

Ordered

Acknowledged

Expected

Shipped, where integrated

Partially Received

Received

Outstanding

PAGE 10 — Sourcing AI, Savings & Procurement Exception Center

Purpose

Provide AI-driven sourcing recommendations and centralized control of commercial procurement risks.

Exception Categories

- Insufficient Supplier Responses
- Supplier Non-Response
- Price Increase
- Quote Above Budget
- Delivery Date Risk
- Supplier Compliance Expiry
- PO Not Acknowledged
- Late PO
- Excessive PO Amendment
- Single-Source Procurement
- Contract Price Variance
- MOQ Exception

AI — SUPPLIER SELECTION RECOMMENDATION

Example:

Required:

1,500 Water Bottles

Needed:

Within 3 Days

Supplier A

Total:

AED 2,895

Lead:

3 Days

Delivery:

96%

Supplier B

Total:

AED 2,760

Lead:

7 Days

Delivery:

84%

Supplier C

Total:

AED 3,120

Lead:

1 Day

Delivery:

98%

TICVAI Recommendation**Supplier A**

Reason:

- Meets required delivery date
- Strong delivery history
- Lower cost than Supplier C
- Supplier B cannot meet required date

Confidence:

91%

Actions:

View Analysis | Select | Compare | Override

AI — TOTAL COST ANALYSIS

TICVAI should consider the true commercial impact.

Example:

Supplier A

Price:

AED 100K

Freight:

AED 8K

Expected Delay Risk:

Low

Supplier B

Price:

AED 94K

Freight:

AED 12K

Expected Delay Risk:

High

Potential emergency purchasing cost if late:

AED 18K

TICVAI may determine Supplier A provides the lower **risk-adjusted total cost**.The system should clearly explain assumptions rather than hiding them.

AI — PRICE BENCHMARKING

TICVAI should compare offered prices against internal historical data.

Example:

Current Quote:

AED 12.80

Last Purchase:

AED 11.40

12-Month Average:

AED 11.65

Contract Reference:

AED 11.80

Variance vs Average:

+9.9%

Alert:

Quotation exceeds recent historical pricing.

Actions**Review History | Negotiate | Accept with Reason**

AI — QUOTE NORMALIZATION

Suppliers may quote in different:

- Currencies
- UOMs
- Pack Sizes
- Freight Terms

Example:

Supplier A:

AED 120 / Case

12 units / Case

= AED 10/unit

Supplier B:

USD 2.60 / Unit

The system should normalize the commercial comparison using approved exchange-rate and UOM services.

AI — AWARD SCENARIO SIMULATION

Before awarding, users should be able to simulate sourcing strategies.

Example:

Scenario A — Supplier A Only

Cost:

AED 180K

Delivery Risk:

Low

Scenario B — Supplier B Only

Cost:

AED 169K

Delivery Risk:

High

Scenario C — Split A/B

Cost:

AED 174K

Delivery Risk:

Medium-Low

Recommendation:

Scenario CThis is especially valuable for large or critical procurement.

AI — SAVINGS ANALYSIS

TICVAI should distinguish different types of savings.

Examples:

Negotiated Savings

Initial Quote → Final Quote

Budget Savings

Budget → Award

Contract Savings

Spot Purchase Benchmark → Contract Price

Avoided Cost

Price Increase Requested → Negotiated Price

This provides more meaningful procurement analytics later in Board 7.

DIRECT PO MANAGEMENT

Not all procurement needs an RFQ.

Direct PO should be permitted when procurement policy allows, for example:

- Existing Contract
- Framework Agreement
- Sole Approved Supplier
- Low-Value Purchase
- Emergency
- Regulated / Proprietary Item

The system should require a reason where competitive RFQ is bypassed.

Example:

Direct Procurement Reason

Existing Contract — SUP-C-1842

Result:

Permitted**SINGLE-SOURCE / SOLE-SOURCE CONTROL**

Where only one supplier is used:

Capture:

- Reason
- Business Justification
- Supplier
- Value
- Supporting Document
- Approval

Examples:

- Proprietary Item
- OEM Requirement
- Exclusive Distributor
- Emergency
- Existing Contract

High-value single-source purchases should be visible to management.

PO PRICE / QUANTITY TOLERANCES

TICVAI should support configurable tolerances that will later be used during receiving and invoice matching.

Example:

Quantity Tolerance

±2%

Price Tolerance

±1%

These controls should travel with the PO so Board 6 can use them during GRN and invoice matching.

TAX, CURRENCY & MULTI-CURRENCY PROCUREMENT

Board 5 shall support procurement in different currencies.

Example:

PO Currency:

USD

Company Base Currency:

AED

Store:

Original Currency

Exchange Rate Reference

Base Currency Equivalent

Exchange-rate ownership should remain within TICVAI's shared Finance / currency service.

F&B PROCUREMENT EXAMPLE

F&B needs:

Beef Patty — 2,000 units.

Board 4:

Approved PR.

↓

Board 5:

RFQ sent to three approved food suppliers.

↓

Quotes received.

↓

TICVAI compares:

Price

Lead Time

Quality

Delivery

MOQ

↓

Procurement selects Supplier A.

↓

Approval.

↓

PO created.

↓

Board 6 later handles:

Receiving + GRN + Invoice Matching**RETAIL PROCUREMENT EXAMPLE**

Event requires:

2,500 Concert T-Shirts.

Board 4 consolidates requirements.



Board 5 RFQ.



Supplier A:

AED 38/unit, 10 days

Supplier B:

AED 40/unit, 5 days

Event:

7 days away.

TICVAI highlights:

Supplier A cannot meet requirement date.

Supplier B recommended despite higher unit price.

This is exactly why procurement intelligence must consider **business need, not price alone.**

EMPLOYEE APP INTEGRATION

Authorized procurement employees should be able to:

- Review RFQ
- View Supplier Responses
- Compare Quotations
- Add Evaluation
- Review Award
- Approve / Reject
- View Purchase Orders
- Check PO Status
- Receive Sourcing Alerts

Complex quotation comparison should remain optimized for desktop, but mobile approval should be supported.

NOTIFICATIONS

Generate notifications for:

- RFQ Issued
- RFQ Closing Soon
- Supplier Quote Received
- Supplier Declined
- Evaluation Required
- Award Approval Required
- Award Approved
- PO Issued
- Supplier Acknowledgement Due
- PO Delivery Delayed
- PO Amendment Approval

APPROVAL ENGINE INTEGRATION

Board 5 shall use TICVAI's common Approval Engine.

Possible approvals:

- Single-Source Procurement
- Sourcing Award
- High-Value PO

- PO Above PR Value
- Budget Exception
- PO Amendment
- Price Above Contract
- Emergency Purchase

Example

Award:

AED 420,000

↓

Procurement Director

↓

Finance

↓

Business Director

↓

Approved

Approval workflows remain tenant-configurable.

SEGREGATION OF DUTIES

The system should support controls such as:

Requester ≠ Buyer

Buyer ≠ Award Approver

Supplier Creator ≠ Supplier Approver

PO Creator ≠ Final PO Approver

depending on the customer's internal control policy.

AUDIT & VERSION CONTROL

Board 5 shall retain full history for:

RFQ

- Creation
- Supplier Invitations
- Changes
- Clarifications
- Closing

Quotations

- Original
- Revisions
- Negotiated Offers

Evaluation

- Score
- User
- Comments
- Changes

Award

- Recommendation
- Approval

- Override Reason

Purchase Order

- Creation
- Amendments
- Supplier Acknowledgement
- Status

No commercial procurement record should be silently overwritten.

BOARD 5 — FINAL 10 SCREENS

#	Screen
1	Sourcing & Purchase Order Command Center
2	RFQ Creation & Sourcing Strategy
3	Supplier Invitation & RFQ Response Management
4	Supplier Quotation Management
5	Quotation Comparison & Supplier Evaluation
6	Negotiation & Best-and-Final-Offer Management
7	Award Recommendation & Approval
8	Purchase Order Creation & Management
9	PO Amendment, Supplier Acknowledgement & Order Tracking
10	Sourcing AI, Savings & Procurement Exception Center

COMPLETE BOARD 5 FLOW

Board 4 Approved Procurement Requirement

↓

Determine Sourcing Route

Existing Contract / Direct PO?

YES

→ Direct PO controls

NO

→ RFQ

↓

Select Eligible Suppliers

↓

Issue RFQ

↓

Supplier Responses

↓

Quotation Validation

↓

Normalize:

Currency + UOM + Freight + Terms

Commercial / Technical Evaluation



Supplier Comparison



Negotiation / BAFO where required

**AI Recommendation + Buyer Evaluation**

Award Recommendation



Approval



Purchase Order



Supplier Acknowledgement



Delivery Schedule



Track Outstanding Quantity

**Next Board****Receiving → GRN → Invoice Matching → PO Closure****ARCHITECTURAL REQUIREMENT**

Board 5 should own the **commercial sourcing and Purchase Order lifecycle**, but it should not duplicate supplier onboarding from Board 4 or warehouse receiving from Board 2.

The correct architecture is:

Board 4 — Procurement Initiation**Supplier + PR + Internal Stock Check + Approval + Demand Consolidation****Board 5 — Sourcing & Commercial Commitment****RFQ + Quotation + Evaluation + Negotiation + Award + Purchase Order****Board 6 — Procurement Receiving & Financial Matching****Supplier Delivery + Receipt + GRN + Inspection + Invoice + Three-Way Match + Discrepancy + Closure**

The most important principle for this Board is:

TICVAI should recommend the best procurement decision based on total business value—not automatically choose the cheapest quotation.

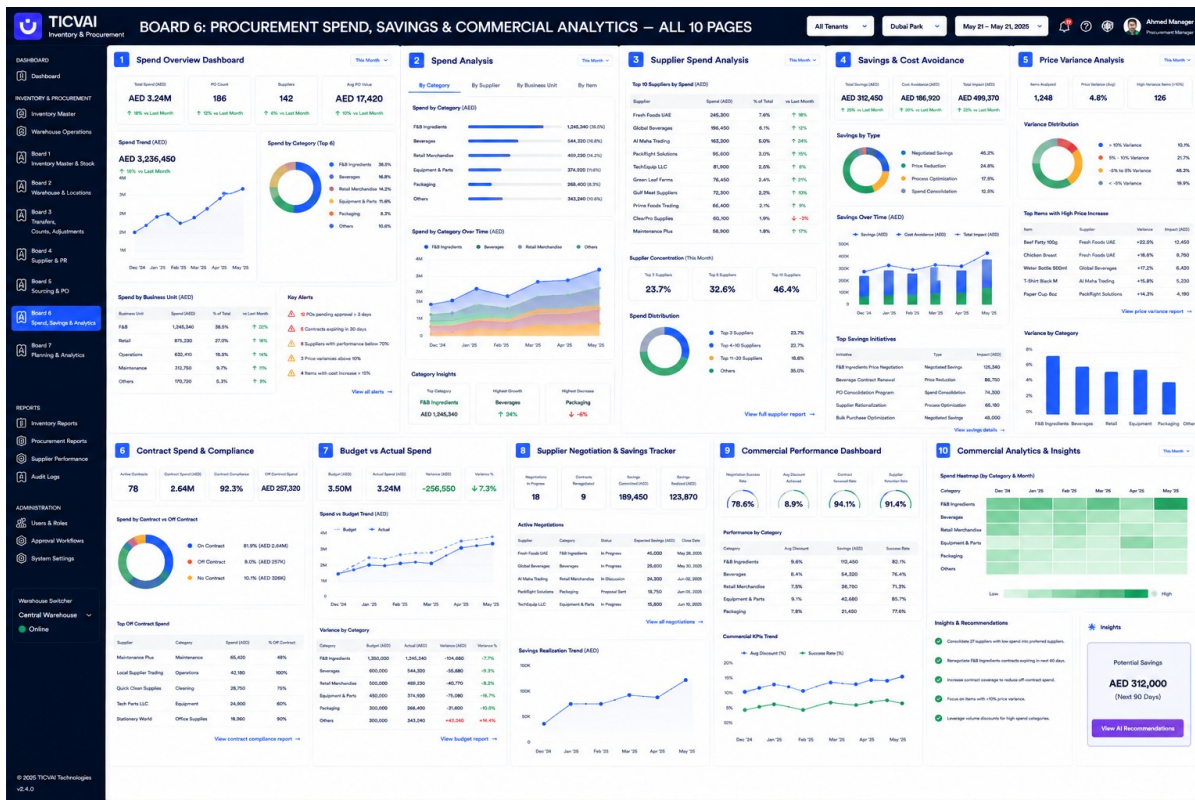
Price matters, but so do:

Delivery → Quality → Compliance → MOQ → Reliability → Required Date → Total Cost → Operational Risk.

This is particularly important because procurement is directly connected to the rest of TICVAI: a cheaper supplier that delivers after an event, after a restaurant stock-out, or after a high-demand retail period can ultimately cost the customer far more than the difference in unit price.

TICVAI INVENTORY & PROCUREMENT — BOARD 6

Receiving, GRN, Invoice Matching & Procurement Closure — Scope of Work



1. Purpose

The Receiving, GRN, Invoice Matching & Procurement Closure Board shall provide TICVAI with the centralized control layer for completing the procurement lifecycle after a Purchase Order has been issued.

Board 5 established:

RFQ → Supplier Quotation → Evaluation → Award → Purchase Order → Supplier Acknowledgement

Board 6 shall establish:

What did the supplier actually deliver? → Was the correct quantity and quality received? → What inventory should become available? → Does the supplier invoice match the PO and receipt? → Are discrepancies within tolerance? → Can the transaction be financially cleared and the PO closed?

The core architecture shall follow:

Purchase Order → Supplier Delivery → Receiving → Inspection → GRN → Inventory Update → Supplier Invoice → 2-Way / 3-Way Match → Exception Resolution → Finance → PO Closure

This Board shall integrate directly with:

Procurement + Warehouse + Shared Inventory + Finance + Supplier Management + Approval Engine + Employee App + AI

A key architectural principle is that Board 6 must **not duplicate the physical warehouse functionality already designed in Board 2**. Board 2 executes the physical receiving and put-away operation; Board 6 controls the **procurement receipt, GRN, matching, financial handoff and PO closure**.

PAGE 1 — Procurement Receiving & Matching Command Center

Purpose

Provide Procurement, Warehouse, Inventory and Finance teams with a consolidated real-time view of supplier deliveries, GRNs, invoice matching, discrepancies and PO closure.

Global Filters

- Tenant
- Company
- Business Unit
- Park
- Venue
- Warehouse
- Supplier
- Procurement Category
- Buyer
- PO Status
- GRN Status
- Invoice Match Status
- Exception Type
- Date / Period

KPI Cards

- Deliveries Expected Today
- Deliveries Received Today
- Late Deliveries
- GRNs Pending
- GRNs Completed
- Inspection Pending
- Invoices Received
- Fully Matched Invoices
- Match Exceptions
- Quantity Variances
- Price Variances
- Value Pending Finance
- POs Ready to Close
- Critical Exceptions

Procurement Completion Pipeline

PO Issued

- Supplier Confirmed
- Expected Delivery
- Received
- Inspection
- GRN
- Invoice
- Match
- Finance

→ Closed

Example:

Expected Deliveries: **26**

Received: **18**

Inspection Pending: **5**

GRN Pending: **7**

Invoices Pending Match: **12**

Match Exceptions: **4**

Ready for Finance: **14**

Ready to Close: **9**

Alerts

Examples:

PO-18438 expected delivery is 6 hours overdue.

GRN-18422 contains a 6.4% quantity variance.

INV-48219 price exceeds PO tolerance by AED 8,400.

Supplier delivered an expired F&B batch.

PO-18427 is fully received and financially matched but remains open.

Quick Actions

Receive Delivery | Create GRN | Inspect Goods | Match Invoice | Resolve Exception | Close PO

PAGE 2 — Expected Deliveries & Receiving Schedule

Purpose

Provide visibility into upcoming supplier deliveries and allow receiving teams to prepare warehouse capacity, staff and dock resources.

Expected Delivery List

Display:

- PO Number
- Supplier
- Delivery Reference
- Expected Date
- Expected Time
- Warehouse
- Dock
- Item Count
- Expected Units
- Priority
- Delivery Status

Delivery Status

- Scheduled
- Confirmed
- Arriving
- Arrived
- Receiving
- Delayed

- Completed
- Cancelled

Calendar Views

Support:

Day | Week | Month | Timeline

Delivery Appointment

Capture:

- Supplier
- PO
- Delivery Date
- Time Slot
- Vehicle
- Driver
- Dock
- Contact
- Special Requirements

Capacity Check

Before confirming a delivery, TICVAI should consider:

- Dock Availability
- Receiving Capacity
- Warehouse Capacity
- Cold Storage Capacity
- Required Employees
- Existing Delivery Schedule

Resource Integration

Example:

10 supplier deliveries expected between:

09:00–11:00

Available receiving teams:

2

Projected requirement:

4

TICVAI Alert:

Receiving capacity is insufficient for the scheduled inbound volume.

Recommendation:

Reschedule two deliveries or allocate additional staff.

PAGE 3 — Supplier Delivery & PO Receiving

Purpose

Record the supplier's actual delivery against one or multiple Purchase Orders.

Delivery Identification

Support:

- PO Scan
- Supplier Delivery Note

- ASN where supported
- Barcode
- QR
- Manual PO Search

Receiving Header

Capture:

- Receipt Number
- Supplier
- PO
- Delivery Note
- Warehouse
- Receiving Dock
- Arrival Time
- Receiver
- Vehicle
- Seal Number where applicable

Line-Level Comparison

For each PO line display:

Ordered

Previously Received

Outstanding

Delivered Today

Accepted

Rejected

Example

Ordered:

1,000

Previously Received:

400

Outstanding:

600

Delivered:

610

Accepted:

600

Rejected:

10

Partial Delivery

TICVAI shall support:

PO Quantity:

1,000

First Receipt:

400

Second Receipt:

350

Remaining:

250

PO remains:

Partially Received

Over-Delivery

Example:

Outstanding:

500

Supplier Delivered:

525

PO Quantity Tolerance:

+2%

Maximum Permitted:

510

Result:

15 units outside tolerance

Actions:

Reject Excess | Exception Approval | Amend PO

PAGE 4 — Inspection, Quality Control & Acceptance

Purpose

Determine whether delivered goods are acceptable before becoming available inventory or progressing to financial matching.

Inspection Requirement

Inspection rules may depend on:

- Item
- Category
- Supplier
- Batch
- Value
- F&B Requirement
- High-Risk Supplier
- Random Quality Control
- Contract

Inspection Types

- Quantity Inspection
- Visual Inspection
- Quality Inspection
- Temperature Check
- Expiry Check
- Batch Verification
- Serial Verification
- Specification Check
- Packaging Check

- Damage Inspection

Inspection Record

Capture:

- Receipt
- Item
- Batch / Serial
- Quantity
- Inspector
- Inspection Type
- Result
- Notes
- Photo
- Supporting Document

Inspection Results

- Accepted
- Accepted with Exception
- Rejected
- Quarantine
- Damaged
- Reinspection Required

F&B Example

Fresh Milk

Required Temperature:

2°C-5°C

Received Temperature:

8°C

Result:

Quality Exception

Actions:

Reject | Quarantine | Supervisor Review

Retail Example

Concert T-Shirt

Ordered:

500

Received:

500

Inspection finds:

24 damaged units

Accepted:

476

Rejected:

24

Only accepted inventory should become usable stock.

PAGE 5 — Goods Receipt Note (GRN) Management

Purpose

Create the formal procurement receipt confirming what the organization accepted from the supplier.

The GRN shall be the critical bridge between:

Purchase Order

and

Inventory / Finance

GRN Header

Capture:

- GRN Number
- Receipt Number
- PO Number
- Supplier
- Warehouse
- Delivery Note
- Receipt Date
- GRN Date
- Receiver
- Status

GRN Lines

For each line:

- Inventory Item
- PO Quantity
- Previously Received
- Delivered
- Accepted
- Rejected
- GRN Quantity
- UOM
- Batch
- Serial
- Expiry
- Unit Cost Reference

GRN Status

- Draft
- Inspection Pending
- Pending Approval
- Posted
- Partially Posted
- Reversed
- Cancelled

Posting GRN

Once approved and posted:

Accepted Quantity

Shared Inventory Ledger



Correct Inventory Status



Warehouse Put-Away where required



PO Received Quantity Updated

Example

PO:

1,000 units

GRN 1:

400

GRN 2:

350

Total Received:

750

Outstanding:

250**GRN Reversal**

Posted GRNs should not simply be deleted.

If an error occurs:

Original GRN

Authorized Reversal



Corrected GRN

with complete audit history.

PAGE 6 — Supplier Invoice Capture & Validation

Purpose

Capture supplier invoices and validate them before financial processing.

Invoice Capture Methods

Support:

- Manual Entry
- PDF Upload
- Structured Electronic Invoice
- API Integration
- Finance / ERP Integration

AI-assisted extraction may be used for uploaded invoices, but extracted values should be validated before posting.

Invoice Header

Capture:

- Supplier Invoice Number
- Supplier

- Invoice Date
- PO Number
- GRN Reference
- Currency
- Subtotal
- Discount
- Freight
- Tax
- Other Charges
- Total
- Payment Terms
- Due Date

Invoice Lines

Capture:

- Item
- Quantity
- UOM
- Unit Price
- Tax
- Discount
- Line Total
- PO Reference
- GRN Reference

Duplicate Invoice Detection

TICVAI should check:

- Supplier
- Invoice Number
- Invoice Date
- Amount
- PO
- Document Fingerprint where supported

Example:

Potential duplicate invoice detected: INV-48219 was previously submitted on 18 Aug.

Actions:

View Existing | Reject Duplicate | Continue with Reason

PAGE 7 — Two-Way / Three-Way Matching

Purpose

Automatically compare procurement documents before financial settlement.

Two-Way Match

Purchase Order

vs

Supplier Invoice

Used where receiving confirmation is not required by policy.

Three-Way Match

Purchase Order

vs

GRN

vs

Supplier Invoice

This should be the primary control for inventory procurement.

Matching Dimensions

Compare:

- Item
- Quantity
- UOM
- Unit Price
- Discount
- Tax
- Freight
- Total
- Currency
- PO
- GRN

Example — Full Match

PO:

100 units × AED 20

GRN:

100 units

Invoice:

100 units × AED 20

Result:

MATCHED

Quantity Exception

PO:

100

GRN:

95

Invoice:

100

Result:

Quantity Mismatch — 5 units

Price Exception

PO:

AED 20

Invoice:

AED 21

Tolerance:

1%

Difference:

5%

Result:

Price Exception

Match Status

- Fully Matched
 - Matched Within Tolerance
 - Quantity Exception
 - Price Exception
 - Tax Exception
 - Freight Exception
 - Missing GRN
 - Missing PO
 - Duplicate Invoice
 - Manual Review
-

PAGE 8 — Match Exception & Dispute Resolution

Purpose

Provide a controlled workflow for resolving procurement and invoice discrepancies.

Exception Types

Quantity

- Invoice > GRN
- Invoice < GRN
- GRN > PO
- Short Delivery
- Over Delivery

Price

- Invoice Price > PO
- Incorrect Discount
- Contract Price Difference

Commercial

- Unexpected Freight
- Unexpected Charge
- Tax Difference
- Currency Difference

Document

- Missing PO
- Missing GRN
- Duplicate Invoice
- Invalid Supplier Invoice

Quality

- Rejected Goods Invoiced
- Damaged Goods Invoiced

- Quarantined Stock Invoiced

Resolution Actions

- Accept Within Tolerance
- Request Supplier Credit Note
- Request Corrected Invoice
- Amend PO
- Adjust GRN where legitimately required
- Reject Invoice
- Partial Approval
- Escalate
- Create Supplier Dispute

Example

Invoice:

AED 52,000

Matched:

AED 48,000

Disputed:

AED 4,000

Action:

Approve AED 48,000 + Hold AED 4,000

where Finance policy supports partial processing.

Dispute Record

Capture:

- Supplier
- Invoice
- PO
- GRN
- Disputed Amount
- Reason
- Owner
- Open Date
- Communication
- Resolution
- Credit Note Reference
- Status

PAGE 9 — Finance Handoff, PO Closure & Supplier Performance

Purpose

Complete the procurement lifecycle once receiving and financial matching conditions are satisfied.

Finance Handoff

Once matched:

PO •

GRN •

Invoice

Approved Match

**Finance / Accounts Payable**

Board 6 should provide Finance with the validated procurement package.

Finance should remain responsible for:

- Accounting Posting
- AP Ledger
- Payment Processing
- Payment Run
- Bank Execution

Board 6 should **not recreate Accounts Payable**.**Finance Status Returned to Procurement**

Display:

- Sent to Finance
- Accepted by Finance
- Payment Scheduled
- Paid
- Finance Exception

PO Closure Conditions

A PO may close when:

- ✓ Required Goods Received
- ✓ GRNs Posted
- ✓ Rejected Quantities Resolved
- ✓ Invoice Matching Complete
- ✓ Outstanding Commitments Resolved
- ✓ Finance Handoff Complete

Closure Types

- Fully Closed
- Short Closed
- Cancelled Balance
- Closed with Approved Variance

Short Close Example

PO:

1,000 units

Received:

980

Supplier confirms:

No further delivery.

Remaining:

20

Authorized action:

Short Close 20 Units

Reason required.

Supplier Performance Update

After completion, automatically update:

- Delivery Timeliness
- Quantity Accuracy
- Quality Acceptance
- Price Accuracy
- Invoice Accuracy
- Lead-Time Accuracy

This feeds the Supplier Performance profile from Board 4.

PAGE 10 — AI Procurement Completion & Exception Center

Purpose

Provide AI-assisted monitoring across deliveries, GRNs, invoice matching and procurement closure.

AI Categories

- Delivery Risk
 - Quantity Variance
 - Quality Risk
 - Invoice Anomaly
 - Match Exception
 - Duplicate Invoice Risk
 - Supplier Performance Risk
 - PO Closure Delay
 - Procurement Leakage
 - Process Bottleneck
-

AI — DELIVERY RISK PREDICTION

TICVAI should identify likely late deliveries before they become operational problems.

Example:

PO-18438

Supplier:

Fresh Foods UAE

Required:

Tomorrow 08:00

Supplier Historical On-Time:

82%

Current Confirmation:

Not Received

Similar Deliveries:

Average 1.4 days late

Inventory Cover:

0.8 days

AI Assessment

High Stock-Out Risk

Recommendation:

1. Contact supplier immediately.

2. Check internal transfer availability.
3. Identify approved alternative supplier.

This should connect back to Board 3 and Board 4 rather than creating duplicate functionality.

AI — RECEIPT ANOMALY DETECTION

Example:

Supplier historically delivers:

98–101% of ordered quantity.

Current Delivery:

112%

AI Alert:

Delivery quantity is materially above normal supplier behavior and exceeds PO tolerance.

Recommended:

Review before acceptance.

AI — QUALITY TREND DETECTION

Example:

Supplier:

Fresh Foods UAE

Rejected Quantity:

Last 30 Days:

1.2%

Current Month:

4.8%

Primary Cause:

Temperature Non-Compliance

AI Alert:

Supplier rejection rate has increased materially.

Recommendation:

Supplier Quality Review

This should feed Board 4 supplier performance.

AI — INVOICE EXTRACTION & VALIDATION

For uploaded supplier invoices, AI may extract:

- Supplier
- Invoice Number
- Invoice Date
- PO
- Item
- Quantity
- Unit Price
- Tax
- Total

Example:

AI Extracted

Invoice:

INV-48219

PO:

PO-18438

Total:

AED 48,420

Confidence:

98%

The user should be able to:

Confirm | Correct | Review Source

Low-confidence fields should be highlighted.

AI — MATCH EXCEPTION ROOT-CAUSE ASSISTANT

Example:

Invoice Exception

Invoice Quantity:

500

GRN:

476

Difference:

24

AI Analysis

Inspection record shows:

24 units rejected as damaged.

Recommended Resolution:

Request supplier credit note for 24 units.

Actions:

Create Dispute | Request Credit Note | Review Inspection

AI — DUPLICATE INVOICE DETECTION

Example:

New Invoice:

INV-84829

Supplier:

Global Beverages

Amount:

AED 42,800

AI identifies another invoice:

INV-84892

Same Supplier

Same PO

Same Amount

Same Date

Similar Document Content

Risk:

Possible Duplicate

The invoice should be held for review rather than automatically rejected.

AI — PO CLOSURE RECOMMENDATION

TICVAI should identify POs that remain operationally open unnecessarily.

Example:

PO-18427

Ordered:

1,000

Received:

1,000

GRN:

Posted

Invoice:

Fully Matched

Finance:

Accepted

Outstanding:

0

Status:

Open for 9 days

Recommendation:

Close PO

Actions:

Close | Review | Keep Open with Reason

AI — PROCUREMENT LEAKAGE DETECTION

TICVAI should identify where actual procurement cost differs materially from approved or negotiated commercial terms.

Example:

Award Value:

AED 184,000

Final Invoiced:

AED 196,400

Difference:

+AED 12,400 / +6.7%

Reasons identified:

Freight:

AED 6,200

Price Variance:

AED 4,800

Additional Charges:

AED 1,400

Recommendation:

Review commercial leakage before final closure.

DELIVERY → WAREHOUSE INTEGRATION

This boundary is important.

Board 6

Controls:

PO → Expected Delivery → Procurement Receipt → GRN**Board 2**

Executes:

Dock → Scan → Inspection → Put-Away → Bin

Example:

Supplier arrives.

↓

Board 6 identifies:

PO-18438

↓

Board 2 executes physical receiving.

↓

Accepted quantities and inspection results return to Board 6.

↓

Board 6 creates / posts:

GRN-18422

↓

Board 1 Shared Inventory Ledger updates.

There should be **one physical receipt transaction**, not duplicate receiving entries in Board 2 and Board 6.

INVENTORY STATUS DURING RECEIVING

Not every received item should immediately become **Available**.

Support:

Received

→ Inspection Pending

→ Quarantine

→ Accepted

→ Available

or

→ Rejected

→ Supplier Return

Example

500 units received.

450 Accepted

30 Inspection Pending

20 Rejected

Shared inventory should show each status separately.

BATCH / LOT / EXPIRY RECEIVING

Particularly important for F&B.

Capture:

- Batch
- Manufacturing Date where applicable
- Expiry Date
- Best Before
- Quantity
- Supplier Batch

- TICVAI Batch Reference

Shelf-Life Rule

Example:

Minimum remaining shelf life:

60 days

Received:

24 days

Result:

Reject / Exception Approval

depending on policy.

This must integrate with FEFO rules already designed in Inventory and F&B.

SERIALIZED RECEIVING

For serialized equipment or merchandise:

PO:

10 Scanners

Supplier delivers:

10

Each serial must be captured:

SER-001

SER-002

SER-003

...

Quantity alone is not sufficient.

TICVAI should prevent duplicate serial registration.

RETURNS TO SUPPLIER

Rejected goods may generate:

Return to Supplier

Capture:

- Return Number
- Supplier
- PO
- GRN
- Item
- Batch / Serial
- Quantity
- Reason
- Dispatch Reference
- Expected Credit Note

Return Lifecycle

Requested

→ Approved

→ Prepared

→ Dispatched

→ Supplier Confirmed

→ Credit Note Pending

→ Resolved

The warehouse physical outbound execution can use Board 2.

CREDIT NOTE MANAGEMENT

Where an invoice has already been submitted or processed, supplier corrections may require a credit note.

Capture:

- Credit Note Number
- Supplier
- Original Invoice
- PO
- GRN
- Amount
- Reason
- Status

Match:

Original Invoice

–

Credit Note

=

Adjusted Payable Amount

Finance remains responsible for accounting treatment.

TOLERANCE MANAGEMENT

TICVAI should support configurable tolerances.

Quantity

Example:

±2%

Price

Example:

±1%

Invoice Total

Example:

AED 50 absolute tolerance

Delivery

Example:

±1 day

Rules may vary by:

- Supplier
- Category
- Item
- Contract
- Company
- Value

Within Tolerance

May auto-clear according to policy.

Outside Tolerance

Requires exception resolution / approval.

PARTIAL RECEIPT & PARTIAL INVOICE

TICVAI must support real procurement scenarios.

Example:

PO

1,000 units

Delivery 1

400

Invoice:

400

Delivery 2

350

Invoice:

350

Remaining

250

The PO remains open.

The system should maintain:

Ordered → Received → Invoiced → Outstanding

at line level.

MULTIPLE GRNs AGAINST ONE PO

Example:

PO-18421

↓

GRN-01:

400

GRN-02:

350

GRN-03:

250

↓

Total:

1,000

↓

PO Fully Received

Similarly, one invoice may reference multiple GRNs where supplier billing arrangements permit.

ONE INVOICE AGAINST MULTIPLE POS

Where policy permits:

Supplier Invoice:

INV-18429

References:

PO-18420

PO-18421

PO-18422

TICVAI should allocate invoice lines to the correct PO and GRN records before matching.

F&B EXAMPLE

Purchase Order

Fresh Milk:

500 units



Supplier Delivery:

500



Temperature Inspection:

450 Accepted

50 Rejected



GRN:

450



Shared Inventory:

450 Inspection Passed / Available after put-away



Supplier Invoice:

500 units



Three-Way Match:

Quantity Exception — 50



AI identifies rejected quantity.



Request:

Credit Note / Corrected Invoice



Supplier resolves.



Finance.



PO Closure.

RETAIL EXAMPLE

PO

Concert T-Shirt:

2,500



Delivery:

2,500



Inspection:

2,476 Accepted

24 Damaged



GRN:

2,476



Invoice:

2,500



Exception:

24 units



Return to Supplier / Credit Note



Invoice Matched



Finance



PO Closed.

EMPLOYEE APP INTEGRATION

Warehouse / Procurement employees should be able to:

- View Expected Deliveries
- Scan PO
- Receive Delivery
- Scan Item
- Record Quantity
- Scan Batch
- Scan Serial
- Record Expiry
- Capture Temperature
- Report Damage
- Capture Photo
- Submit Inspection
- Create Receipt
- View GRN
- Report Exception

Managers should be able to:

- Approve GRN Exception
- Approve Over-Delivery
- Review Match Exception
- Approve Short Closure

Complex invoice matching should remain optimized for desktop.

APPROVAL ENGINE INTEGRATION

Board 6 shall use the shared TICVAI Approval Engine.

Potential approval events:

- Over-Delivery
- Receipt Outside Tolerance

- Quality Exception Acceptance
- GRN Reversal
- Price Variance
- Invoice Variance
- Manual Match
- Supplier Return
- PO Short Close

Example

Invoice Variance:

AED 250

→ Auto-clear where policy permits.

Invoice Variance:

AED 12,000

→ Procurement Manager + Finance.

FINANCE INTEGRATION

This Board should connect very closely to the Finance module we already designed.

The boundary should be:

Inventory & Procurement

Owns:

PO + GRN + Match + Procurement Exception

Finance

Owns:

Accounting + AP + Payment + Bank + Financial Reporting

Therefore:

Matched Procurement Package

↓

Finance

↓

AP Posting

↓

Payment

↓

Payment Status returned to Procurement

This prevents SoftLab from implementing a second Accounts Payable engine inside Procurement.

AUDIT & TRACEABILITY

Every procurement completion transaction shall retain:

- PO
- Supplier
- Delivery
- Receipt
- GRN
- Item
- Batch / Serial
- Quantity

- Inspection
- Invoice
- Match Result
- Exception
- Approval
- User
- Device
- Date / Time
- Previous State
- New State

Management should be able to trace:

PR

- RFQ
- Quote
- Award
- PO
- Supplier Delivery
- Receipt
- Inspection
- GRN
- Inventory
- Invoice
- Match
- Finance
- Payment Status
- PO Closure

This gives TICVAI a complete **procure-to-pay audit chain**.

BOARD 6 — FINAL 10 SCREENS

#	Screen
1	Procurement Receiving & Matching Command Center
2	Expected Deliveries & Receiving Schedule
3	Supplier Delivery & PO Receiving
4	Inspection, Quality Control & Acceptance
5	Goods Receipt Note (GRN) Management
6	Supplier Invoice Capture & Validation
7	Two-Way / Three-Way Matching
8	Match Exception & Dispute Resolution
9	Finance Handoff, PO Closure & Supplier Performance
10	AI Procurement Completion & Exception Center

COMPLETE BOARD 6 FLOW

PURCHASE ORDER

Board 5 PO Issued



Supplier Acknowledgement



Delivery Schedule



Expected Delivery



Supplier Arrives



PHYSICAL RECEIVING

Board 2 executes:

Dock → Scan → Quantity → Inspection



ACCEPTANCE

Accepted

→ GRN

→ Inventory

→ Put-Away

Rejected

→ Quarantine / Return to Supplier



INVOICE

Supplier Invoice



Capture / AI Extract



Validation



THREE-WAY MATCH

PO

vs

GRN

vs

Invoice



MATCH?

YES

→ Finance Handoff

WITHIN TOLERANCE?

YES

→ Auto-Clear according to policy

NO

→ Exception

→ Investigation

→ Supplier Dispute / Credit Note / PO Amendment / Authorized Resolution

↓

Approved Match

↓

Finance / AP

↓

Payment Status

↓

PO Closure

↓

Supplier Performance Updated

↓

Audit Complete

COMPLETE PROCUREMENT ARCHITECTURE — BOARDS 1-6

At this point, the architecture becomes very clean:

BOARD 1 — INVENTORY MASTER & STOCK LEDGER

Item → SKU → UOM → Batch → Serial → Cost → Stock → Reorder

↓

BOARD 2 — WAREHOUSE EXECUTION

Warehouse → Zone → Bin → Receive → Put-Away → Pick → Pack → Dispatch

↓

BOARD 3 — INVENTORY CONTROL

Transfer → Count → Variance → Adjustment → Reconciliation

↓

BOARD 4 — PROCUREMENT INITIATION

Supplier → Demand → Internal Stock Check → PR → Approval → Consolidation

↓

BOARD 5 — SOURCING & COMMERCIAL COMMITMENT

RFQ → Quote → Compare → Negotiate → Award → PO

↓

BOARD 6 — PROCUREMENT COMPLETION

Delivery → Inspection → GRN → Invoice → Match → Finance → Closure

Key recommendation.

I would make one requirement especially explicit in the final SOW:

TICVAI shall implement one continuous Procure-to-Pay transaction chain with shared identifiers and audit history, rather than separate PR, RFQ, PO, warehouse, GRN and invoice systems connected only by manual references.

For example, from any invoice, an authorized user should be able to navigate backward:

Invoice → GRN → Receipt → PO → Award → Supplier Quote → RFQ → Purchase Request → Original Demand

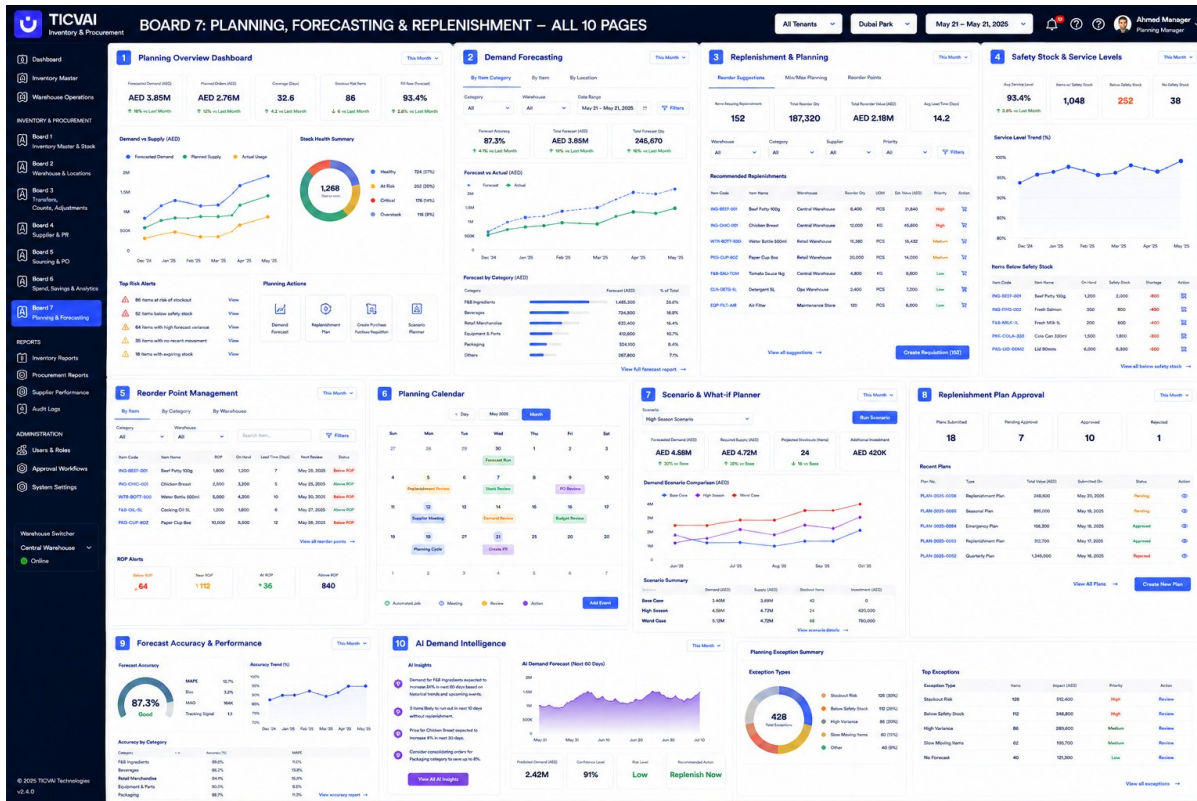
And forward:

PR → RFQ → Award → PO → Delivery → GRN → Invoice → Finance Status → Closure

That is important for TICVAI because it turns Inventory & Procurement into one integrated enterprise engine rather than several disconnected backend modules.

TICVAI INVENTORY & PROCUREMENT — BOARD 7

Planning, Analytics, AI Optimization, Controls & Integration — Scope of Work



1. Purpose

Board 7 shall provide the **management, intelligence, optimization and governance layer** across the complete TICVAI Inventory & Procurement ecosystem.

Boards 1–6 manage the operational lifecycle:

Inventory Master → Warehouse → Transfers & Counts → Procurement Demand → Sourcing & PO → Receiving & Matching

Board 7 shall answer the management questions:

What will we need? → When will we need it? → Where will shortages occur? → Are we carrying too much stock? → How efficiently are we purchasing? → Which suppliers are performing? → Where are we losing money? → What should TICVAI recommend or automate?

The Board shall consolidate data from:

- Inventory
- Warehouse
- F&B
- Retail
- Rental
- Resource Management
- Procurement
- Suppliers
- Finance
- Ticketing / Attendance
- Events

- Historical Sales
- Reservations
- Weather/external data where integrated
- Approval Engine
- Audit Engine

The intelligence architecture shall follow:

Operational Data → Forecast → Demand Plan → Inventory Position → Procurement Plan → AI Recommendation → Approval / Action → Outcome → Continuous Learning

PAGE 1 — Inventory & Procurement Executive Command Center

Purpose

Provide senior management with a consolidated real-time view of inventory, warehouse and procurement performance across TICVAI.

Global Filters

- Tenant
- Company
- Business Unit
- Park
- Venue
- Warehouse
- Department
- Category
- Supplier
- Period
- Comparison Period

Executive KPI Cards

- Total Inventory Value
- Available Inventory Value
- Inventory Turnover
- Days of Inventory
- Inventory Accuracy %
- Stock-Out Rate
- Overstock Value
- Slow-Moving Value
- Dead Stock Value
- Expiry Risk Value
- Shrinkage %
- Open PR Value
- Open PO Value
- Procurement Spend
- Savings
- Supplier On-Time Delivery %
- Purchase Price Variance
- Critical Risks

Enterprise Inventory Health

Example:

Area	Inventory	Stock-Out	Overstock	Accuracy	Health
F&B	AED 1.42M	2.1%	AED 82K	99.1%	Good
Retail	AED 2.85M	3.8%	AED 240K	98.4%	Warning
Operations	AED 940K	0.8%	AED 62K	99.5%	Good
Rental	AED 680K	1.4%	AED 38K	99.0%	Good

Management Alerts

Examples:

AED 420K of inventory is classified as slow-moving.

Retail stock-out exposure for the upcoming event weekend is estimated at AED 86K revenue.

3 suppliers have fallen below required delivery-performance thresholds.

F&B expiry exposure for the next 14 days is AED 18,600.

Open procurement commitments are projected to exceed monthly plan by 7.4%.

Quick Actions

Demand Forecast | Replenishment Plan | Inventory Optimization | Procurement Analytics | Supplier Analytics | AI Recommendations

PAGE 2 — Demand Forecasting & Consumption Planning

Purpose

Forecast future inventory requirements using operational demand rather than relying only on historical averages.

Forecast Inputs

TICVAI should support:

- Historical Consumption
- Retail Sales
- F&B Recipe Consumption
- Ticket Sales
- Reservations
- Attendance Forecast
- Event Schedule
- Membership Activity
- Seasonality
- Day of Week
- Holidays
- Promotions
- Weather where integrated
- Venue Capacity
- Existing Bookings
- Current Inventory
- Open Purchase Orders
- Transfers
- Supplier Lead Time

Forecast Hierarchy

Forecast by:

Enterprise

- Company
- Park
- Venue
- Warehouse
- Outlet
- Category
- Item / SKU

Example — F&B

Upcoming Saturday:

Expected Attendance:

18,500

Historical Burger Conversion:

12.4%

Forecast Burgers:

2,294

Each Burger:

1 Beef Patty

Current Available:

1,420

Incoming:

400

Safety Stock:

300

Projected Shortfall:

774 Patties

Example — Retail

Concert Ticket Sales:

12,800

Expected Merchandise Conversion:

18%

Expected T-Shirt Demand:

2,304

Available:

1,680

Incoming:

400

Projected Shortfall:

224

Forecast Views

Support:

Daily | Weekly | Monthly | Seasonal | Event-Based

Forecast Accuracy

The system should compare:

Forecast vs Actual

and track accuracy over time.

PAGE 3 — Replenishment & Procurement Planning

Purpose

Convert forecast demand and inventory policies into actionable replenishment recommendations.

Planning Inputs

For each item/location:

Current Available

In Transit

Open PO

Expected Production

Reserved

Forecast Demand

Safety Stock

Projected Inventory Position

Recommended Actions

TICVAI may recommend:

- No Action
- Internal Replenishment
- Internal Transfer
- Purchase
- Expedite PO
- Delay PO
- Reduce PO
- Cancel Remaining PO where permitted
- Increase Safety Stock
- Reduce Safety Stock

Planning Table

Item	Available	Demand	Incoming	Safety	Gap	Recommendation
Beef Patty	1,420	2,294	400	300	-774	Purchase
T-Shirt M	1,680	2,304	400	150	-374	Transfer + Purchase
Water 500ml	8,400	5,200	2,000	1,500	+3,700	No Purchase

Planning Horizon

Support:

- Next 24 Hours
- 3 Days
- 7 Days
- 14 Days
- 30 Days
- 90 Days
- Custom

Planner Actions**Accept Recommendation****Modify****Create Transfer****Create Draft PR****Defer****Ignore with Reason****PAGE 4 — Inventory Optimization & Stock Health****Purpose**

Identify inventory that is too high, too low, slow-moving, obsolete or at risk of expiry.

Inventory Health Classifications

- Healthy
- Low Stock
- Critical
- Overstock
- Slow Moving
- Non-Moving
- Dead Stock
- Expiry Risk
- Obsolete

Slow-Moving Analysis

Configurable definitions:

Example:

No movement:

30 Days → Slow

90 Days → Very Slow

180 Days → Dead Stock

Definitions should vary by category.

Fresh food obviously cannot use the same thresholds as merchandise or spare parts.

Overstock Analysis

Example:

Logo Mug

Available:

1,840

Average Monthly Demand:

220

Inventory Cover:

8.4 Months

Target:

2 Months

Excess:

1,400 units

Value:

AED 39,200

Recommended Actions

- Transfer to Higher-Demand Location
- Reduce Reorder Quantity
- Stop Replenishment
- Return to Supplier where allowed
- Promotional Action
- Bundle
- Markdown recommendation to Retail
- Consume / Prioritize in F&B
- Write-Off Review

Expiry Risk

Example:

Fresh Juice:

240 units

Expires:

5 days

Forecast Consumption:

140

Potential Waste:

100 units

Estimated Cost:

AED 1,800

Recommendation:

Redistribute to high-demand outlets.

PAGE 5 — Procurement Spend, Savings & Commercial Analytics

Purpose

Provide management with comprehensive visibility into procurement spend and commercial performance.

KPI Cards

- Total Spend
- PO Value
- Average PO Value
- Savings

- Cost Avoidance
- Purchase Price Variance
- Contract Spend %
- Off-Contract Spend
- Emergency Procurement %
- Single-Source Spend
- Average Procurement Cycle Time

Spend Analysis

Analyze by:

- Supplier
- Category
- Department
- Company
- Park
- Venue
- Buyer
- Item
- Month
- Quarter
- Year

Savings Analysis

Separate:

Negotiated Savings

Budget Savings

Contract Savings

Volume Savings

Cost Avoidance

Example

Annual Procurement Spend:

AED 18.4M

Negotiated Savings:

AED 820K

Contract Savings:

AED 340K

Cost Avoidance:

AED 280K

Total Procurement Value Contribution:

AED 1.44M

Purchase Price Variance

Example:

Water Bottle

Previous:

AED 1.82

Current:

AED 1.95

Change:

+7.1%

Annualized Impact:

+AED 84K

PAGE 6 — Supplier Performance & Risk Analytics

Purpose

Provide centralized supplier intelligence based on actual operational performance.

Supplier Scorecard

Track:

- On-Time Delivery
- Fill Rate
- Quality Acceptance
- Quantity Accuracy
- Price Accuracy
- Invoice Accuracy
- Lead-Time Reliability
- Response Rate
- Dispute Rate
- Compliance
- Contract Status

Example

Fresh Foods UAE

Overall Score:

89 / 100

Delivery:

82%

Quality:

96%

Quantity:

98%

Invoice:

94%

Compliance:

100%

Trend:

Declining

Supplier Ranking

Supplier	Spend	Delivery	Quality	Price	Overall
Supplier A	AED 2.8M	96%	95%	91%	94
Supplier B	AED 1.9M	84%	97%	96%	89
Supplier C	AED 1.2M	98%	98%	82%	92

Supplier	Spend	Delivery	Quality	Price	Overall
----------	-------	----------	---------	-------	---------

Risk Indicators

- Late Delivery Trend
- Quality Decline
- Price Inflation
- Compliance Expiry
- High Dependency
- Single Supplier Dependency
- Repeated Invoice Issues
- Frequent Short Deliveries

Supplier Concentration

Example:

Category:

Fresh Food

Spend with Supplier A:

72%

Risk:

High Dependency

Recommendation:

Qualify an alternative supplier.

PAGE 7 — Warehouse & Inventory Performance Analytics

Purpose

Measure the operational effectiveness of warehouse and inventory-control processes.

Warehouse KPIs

- Receiving Time
- Put-Away Time
- Picking Time
- Picking Accuracy
- Packing Accuracy
- Dispatch SLA
- Warehouse Capacity %
- Dock Utilization
- Task Completion Rate
- Employee Productivity
- Inventory Accuracy
- Count Variance
- Transfer Accuracy

Warehouse Comparison

Warehouse	Accuracy	Pick SLA	Capacity	Errors	Health
Central	99.6%	97%	78%	0.4%	Excellent
F&B	99.1%	94%	86%	0.9%	Good
Retail	98.4%	91%	72%	1.6%	Warning

Bottleneck Analysis

Example:

Receiving:

12 min avg

Put-Away:

42 min avg

Picking:

18 min avg

Packing:

11 min avg

AI identifies:

Put-Away is the primary bottleneck.

Capacity Heatmap

Display utilization by:

Warehouse → Zone → Day → Time

This helps management anticipate congestion before it occurs.

PAGE 8 — Inventory & Procurement Approval / Control Center

Purpose

Provide authorized management with one centralized workspace for inventory and procurement exceptions requiring approval.

This shall consume the **shared TICVAI Approval Engine**, not create another approval framework.

Approval Categories

Inventory

- Stock Adjustment
- Write-Off
- Negative Stock Override
- Large Variance
- Emergency Transfer

Warehouse

- Over-Delivery
- Manual Movement Override
- Quality Exception

Procurement

- Purchase Request
- Single-Source Purchase
- RFQ Award

- High-Value PO
- PO Amendment
- Budget Exception

Receiving / Matching

- GRN Exception
- Invoice Variance
- Manual Match
- Short Close

Approval Inbox

Request	Type	Value	Risk	Waiting	Action
ADJ-1842	Write-Off	AED 18K	High	2h	Review
PO-1848	PO Award	AED 420K	Medium	4h	Review
INV-4821	Price Variance	AED 8.4K	High	1d	Review

SLA

Display:

- Within SLA
- Approaching SLA
- Overdue

Escalation

Approval workflows should support automatic escalation based on TICVAI's common policy engine.

PAGE 9 — Integration, API & Automation Control Center

Purpose

Provide administrators with visibility and control over Inventory & Procurement integrations.

Integration Areas

- Retail
- F&B
- Finance / ERP
- Supplier Systems
- Employee App
- Resource Management
- Event Management
- Ticketing
- CRM where applicable
- External Logistics
- Barcode / RFID Systems
- BI / Data Warehouse

Integration Dashboard

Display:

- Integration
- Direction
- Last Sync

- Transactions
- Success
- Failed
- Pending
- Health

Example:

Integration	Last Sync	Success	Failed	Health
Retail POS	10:32	12,840	2	Healthy
F&B	10:31	8,422	0	Healthy
Finance	10:28	1,840	4	Warning
Supplier API	10:25	620	8	Warning

API Capabilities

Support controlled APIs for:

- Inventory Lookup
- Stock Availability
- Inventory Movement
- Purchase Request
- Purchase Order
- Supplier
- GRN
- Invoice Match Status
- Transfer
- Replenishment Recommendation

Event / Webhook Architecture

Where supported:

Inventory Updated

PO Created

GRN Posted

Stock Below Threshold

Supplier Delivery Received

should be capable of generating platform events for subscribed TICVAI services.

Integration Exception

Example:

Retail POS posted 14 sales while Inventory API was unavailable.

Status:

Queued for Reprocessing

Actions:

Retry | Investigate | View Transactions

PAGE 10 — TICVAI AI Inventory & Procurement Copilot

Purpose

Provide a conversational AI layer across the complete Inventory & Procurement module.

Instead of forcing managers to navigate multiple reports, they should be able to ask TICVAI questions directly.

Example Questions

Which items will run out during the next seven days?

What should I purchase today?

Which warehouses have excess stock?

Which suppliers are performing badly?

Why did F&B food cost increase this month?

Which POs are likely to arrive late?

Show me inventory at risk of expiry.

How much inventory is currently tied up in slow-moving stock?

Why did Retail inventory accuracy fall?

Which procurement decisions saved the most money this quarter?

AI Response Example

Question:

What should I purchase for this weekend?

TICVAI AI

Based on:

- 28,400 expected visitors
- Existing reservations
- Event schedule
- Current inventory
- Open POs
- Transfers
- Safety stock
- Supplier lead times

I identified:

14 items requiring action

Estimated Purchase:

AED 84,600

Internal Transfer Opportunity:

AED 18,200

Urgent Purchase:

5 items

Potential Stock-Out Revenue Exposure:

AED 42,800

Actions:

View Plan | Create Draft PRs | Review Items

AI — DEMAND FORECASTING

The AI engine should combine multiple demand signals.

Traditional

Historical Consumption
•

Operational

Ticket Sales + Reservations + Events
•

Commercial

Promotions + Retail Campaigns
•

Environmental

Weather / Seasonality where integrated
↓

TICVAI Demand Forecast

This is especially valuable for TICVAI because inventory demand at attractions and entertainment venues is heavily influenced by **attendance**.

AI — CROSS-MODULE DEMAND INTELLIGENCE

This is one of the most important capabilities I recommend.

Example:

TICVAI Ticketing sees:

Saturday Attendance Forecast +32%

↓

AI identifies likely impact:

F&B

Water:

+28%

Burgers:

+24%

Ice Cream:

+38%

Retail

Souvenir T-Shirts:

+19%

Caps:

+17%

Operations

Wristbands:

+31%

↓

Inventory checks availability.

↓

AI identifies shortages.

↓

Checks internal redistribution.

↓

Creates recommendations.

↓

Draft PRs where necessary.

This connects the entire TICVAI platform instead of treating Inventory as an isolated ERP feature.

AI — DYNAMIC SAFETY STOCK

Instead of using one fixed safety-stock level forever, TICVAI should recommend adjustments based on:

- Demand Volatility
- Supplier Reliability
- Lead Time
- Season
- Events
- Stock-Out Cost

Example:

Normal Safety Stock:

300

Upcoming Eid period:

Recommended:

520

Reason:

Expected demand increase + supplier holiday lead-time risk.

The user must approve changes unless tenant policy explicitly permits automation.

AI — INVENTORY REBALANCING

Example:

Main Gift Shop

T-Shirt M:

Available:

8

Forecast:

64

Waterpark Shop

Available:

92

Forecast:

30

Event Store

Available:

74

Forecast:

22

AI Recommendation:

Transfer 56 units to Main Gift Shop

rather than purchasing additional stock.

Estimated Purchase Avoidance:

AED 2,352

AI — PROCUREMENT CONSOLIDATION

Example:

Three departments request the same item:

F&B:

500

Retail:

800

Event:

1,200

AI recommends:

Consolidate 2,500 units into one sourcing event.

Potential Volume Saving:

AED 8,400

AI — CASH & WORKING CAPITAL OPTIMIZATION

TICVAI should consider the cost of holding unnecessary inventory.

Example:

Inventory Value:

AED 8.4M

Slow / Excess:

AED 1.2M

Potential Reducible Stock:

AED 640K

AI Recommendation:

Reduce selected reorder quantities and delay 6 non-critical Purchase Orders.

Potential Working Capital Release:

AED 420K

Finance should consume these recommendations, but Finance remains responsible for financial accounting.

AI — EXPLAINABLE RECOMMENDATIONS

Every major AI recommendation should explain:

Recommendation

Purchase 800 units

Why?

- Current Available: 320
- Forecast Demand: 920
- Safety Stock: 200
- Incoming PO: 0
- Supplier Lead Time: 4 days
- Event begins in 6 days

Confidence

93%

Actions

Accept | Modify | Simulate | Ignore

The AI should not behave like a black box.

AI — SIMULATION / WHAT-IF ANALYSIS

Managers should be able to simulate scenarios.

Example:

What happens if attendance is 20% higher than forecast?

TICVAI recalculates:

- F&B Demand
- Retail Demand
- Inventory Shortages
- Procurement Requirements
- Warehouse Workload
- Supplier Capacity
- Working Capital

Another example:

What happens if Supplier A is delayed by 5 days?

TICVAI identifies affected:

Items → Locations → F&B Products → Retail Products → Events

and recommends mitigation.

AI — RECOMMENDATION GOVERNANCE

AI recommendations shall have configurable execution modes.

Level 1 — Insight Only

AI displays information.

Level 2 — Recommendation

AI recommends an action.

Level 3 — Draft Action

AI creates:

- Draft Transfer
- Draft PR
- Draft Replenishment

Level 4 — Auto-Execute Within Policy

Only where explicitly enabled.

Example:

Automatically create internal replenishment below AED 500 equivalent operational value.

Level 5 — Approval Required

AI creates action and routes it through the Approval Engine.

I recommend TICVAI default to **Levels 2-3**, with Level 4 enabled only for clearly defined low-risk workflows.

ALERT & EXCEPTION MANAGEMENT

Board 7 shall consolidate intelligent alerts across all Inventory & Procurement Boards.

Inventory

- Low Stock
- Stock-Out
- Overstock
- Expiry
- Negative Stock
- Shrinkage

Warehouse

- Capacity
- Picking Delay
- Receiving Delay
- Dispatch Delay

Procurement

- PR Delay
- RFQ Delay
- Supplier Risk
- PO Delay
- Price Increase

Receiving

- Quality Failure
- GRN Exception
- Invoice Exception
- Match Failure

Alert Prioritization

AI should avoid flooding users.

Prioritize by:

Financial Impact + Operational Impact + Guest Impact + Urgency + Probability

REPORTING & EXPORT

Board 7 should provide standard reports including:

- Inventory Valuation
- Stock Position
- Stock Movement
- Inventory Aging
- Expiry Report
- Slow-Moving Stock
- Dead Stock
- Shrinkage
- Inventory Accuracy
- Warehouse Performance
- Procurement Spend
- Supplier Spend
- Supplier Performance
- Purchase Price Variance
- Procurement Savings
- PR Cycle Time
- PO Cycle Time
- GRN Performance
- Invoice Match Performance
- Procurement Exceptions

Reports should support:

View → Filter → Drill Down → Export → Schedule

ROLE-BASED DASHBOARDS

Board 7 should not show everyone the same information.

Inventory Manager

Stock Health + Accuracy + Replenishment

Warehouse Manager

Capacity + Tasks + SLA + Accuracy

Procurement Manager

Spend + RFQ + PO + Savings + Supplier

Finance

Inventory Value + Commitments + Variance + Match

Operations Manager

Availability + Operational Risk

Executive

High-Level KPIs + Risk + Savings + Working Capital

AI Assistant

Responses must respect the user's permissions.

A manager should never retrieve restricted data merely by asking the AI.

AUDIT & AI GOVERNANCE

Maintain audit history for:

- Forecast Changes
- Manual Overrides
- AI Recommendations
- Accepted Recommendations
- Rejected Recommendations
- Auto-Created PR
- Auto-Created Transfer
- Approval
- Policy Changes

Example:

AI Recommendation

Transfer:

100 units

User Modified:

80 units

Reason:

Expected local event demand.

This becomes useful feedback for future AI recommendations.

BOARD 7 — FINAL 10 SCREENS

#	Screen
1	Inventory & Procurement Executive Command Center
2	Demand Forecasting & Consumption Planning
3	Replenishment & Procurement Planning
4	Inventory Optimization & Stock Health
5	Procurement Spend, Savings & Commercial Analytics
6	Supplier Performance & Risk Analytics

#	Screen
7	Warehouse & Inventory Performance Analytics
8	Inventory & Procurement Approval / Control Center
9	Integration, API & Automation Control Center
10	TICVAI AI Inventory & Procurement Copilot

COMPLETE BOARD 7 INTELLIGENCE FLOW

DATA

Ticketing

Events

Reservations

Retail

F&B

Inventory

Warehouse

Procurement

Suppliers

Finance

↓

TICVAI INTELLIGENCE LAYER

Analyze:

Demand + Stock + Cost + Lead Time + Supplier Performance + Operational Capacity

↓

Forecast:

What will be needed?

↓

Determine:

Where and when?

↓

Check:

Existing Stock + Transfers + Incoming PO

↓

Recommend:

OPTION 1

No Action

OPTION 2

Replenish Internally

OPTION 3

Transfer Between Locations

OPTION 4

Purchase

OPTION 5

Expedite Existing PO

OPTION 6

Reduce / Delay Procurement

↓

Simulate Impact

↓

Human Review / Approval

↓

Execute through Boards 1–6

↓

Measure Actual Outcome

↓

Improve Future Recommendations

COMPLETE INVENTORY & PROCUREMENT ARCHITECTURE — BOARDS 1–7

Board	Responsibility
Board 1	Inventory Master & Stock Control
Board 2	Warehouse, Location & Fulfilment Operations
Board 3	Transfers, Counts, Adjustments & Reconciliation
Board 4	Supplier & Purchase Request Management
Board 5	RFQ, Sourcing, Quotations & Purchase Orders
Board 6	Receiving, GRN, Invoice Matching & Procurement Closure
Board 7	Planning, Analytics, AI Optimization, Controls & Integration

Architectural recommendation

Board 7 should **not become another transactional system**. It should sit above Boards 1–6 as TICVAI's **Inventory & Procurement Intelligence Layer**.

The architecture should therefore be:

Transactional Engines — Boards 1–6**Unified Inventory & Procurement Data****BOARD 7 — INTELLIGENCE & CONTROL****Forecast**

→ Analyze

→ Detect

→ Recommend

→ Simulate

→ Approve



Send execution back to the correct engine:

Transfer → Board 3**Warehouse Task → Board 2****Purchase Request → Board 4****RFQ / PO → Board 5****Receiving / Match → Board 6**

This prevents AI from becoming a disconnected chatbot. It makes AI an **actionable layer on top of the actual TICVAI workflows**.

The strongest differentiator here is the connection between **ticket demand and inventory/procurement planning**. TICVAI already knows ticket sales, attendance forecasts, events, F&B demand and retail activity. That information should automatically improve inventory forecasts and purchasing decisions.

For an attractions and entertainment platform, that is considerably more valuable than building a traditional standalone inventory system.